



ARISTOTLE UNIVERSITY OF THESSALONIKI

**Contactless ECG Reconstruction from CW
Radar Signals using Morphology-Aware
Architectures and Attention-Guided Neural
Networks**

by

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A thesis submitted in partial fulfillment for the
Undergraduate degree

in the
Faculty of Sciences
School of Informatics

Supervising Professor: Nikolaos Laskaris

September 27, 2026

Abstract

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This thesis investigates the feasibility of reconstructing continuous Electrocardiogram (ECG) waveforms from non-contact Continuous-Wave (CW) radar measurements of mechanical chest displacement using deep learning. A key challenge is the presence of dominant respiratory and low-frequency motion components, which can obscure the much smaller mechanical manifestations of cardiac activity.

The proposed methodology develops from a 1D-CNN baseline toward wavelet- and attention-based architectures. A Maximal Overlap Discrete Wavelet Transform (MODWT) is incorporated to provide a multi-scale representation of the radar signal, while attention mechanisms are investigated to improve the extraction of temporally relevant cardiac information. A morphology-informed fusion mechanism and a spectral-domain loss are additionally explored to improve waveform reconstruction.

The approaches are evaluated using a sample of 10 recordings, 8 for training-validation and 2 for testing, from 30 healthy volunteers under different physiological conditions. Temporal Attention and Local Cross-Attention provide only marginal improvements over the baseline in terms of mean squared error. In contrast, the Morphological Fusion approach reduces the mean test MSE from 0.04374 to 0.03785, corresponding to a 13.4% reduction. This improvement is primarily associated with reduced inter-beat noise, while the reconstructed QRS complexes remain broader than those of the reference ECG and sharp R-peak morphology is not fully recovered.

Overall, the results demonstrate the feasibility of learning a mapping from radar-derived mechanical chest displacement to ECG-related waveforms, while highlighting the gap between global reconstruction error and fine-grained ECG morphological fidelity. The findings provide a basis for further investigation of morphology-aware objectives, richer radar representations, and advanced sequence or generative models.

Περίληψη

Σχολή Θετικών Επιστημών

Τμήμα Πληροφορικής

Πτυχιακή Εργασία

Ανέπαφη Ανακατασκευή Ηλεκτροκαρδιογραφήματος από Σήματα Ραντάρ Συνεχούς Κύματος μέσω Αρχιτεκτονικών με Επίγνωση Μορφολογίας και Νευρωνικών Δικτύων Καθοδηγούμενης Προσοχής

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Η παρούσα πτυχιακή εργασία διερευνά τη μη επεμβατική ανακατασκευή συνεχών σημάτων ηλεκτροκαρδιογραφήματος (ECG) από μετρήσεις ραντάρ συνεχούς κύματος (CW Radar), μέσω τεχνικών βαθιάς μάθησης. Η κύρια πρόκληση έγκειται στην ισχυρή παρεμβολή της αναπνοής, η οποία επισκιάζει τις ανεπαίσθητες μηχανικές δονήσεις της καρδιάς.

Η προτεινόμενη μεθοδολογία εξελίσσει ένα βασικό μοντέλο 1D-CNN, αξιοποιώντας ανάλυση πολλαπλής κλίμακας μέσω του MODWT (Maximal Overlap Discrete Wavelet Transform) και μηχανισμούς προσοχής (attention). Παράλληλα, εισάγεται μια συνάρτηση απώλειας φασματικού πεδίου και ένας μηχανισμός μορφολογικής σύντηξης (morphological fusion) για τη βελτιστοποίηση της παραγόμενης κυματομορφής.

Η πειραματική αξιολόγηση σε ένα δείγμα 10 εγγραφών (8 για εκπαίδευση-επικύρωση και 2 για δοκιμή) από 30 εθελοντές υπό διάφορες φυσιολογικές συνθήκες έδειξε ότι, ενώ οι αρχιτεκτονικές Temporal και Local Cross-Attention προσφέρουν οριακές βελτιώσεις, η προσέγγιση Morphological Fusion μείωσε το μέσο τετραγωνικό σφάλμα (MSE) από 0.04374 σε 0.03785 (βελτίωση 13.46%). Ωστόσο, η βελτίωση αυτή προήλθε κυρίως από την εξομάλυνση του θορύβου μεταξύ των παλμών, καθώς τα ανακατασκευασμένα συμπλέγματα QRS παρέμειναν μορφολογικά ευρύτερα, αδυνατώντας να αναπαράγουν πλήρως την αιχμηρότητα των κορυφών R.

Συμπερασματικά, η εργασία αποδεικνύει ότι, αν και είναι εφικτή η αντιστοίχιση της μηχανικής κίνησης του θώρακα με ηλεκτρικά καρδιακά σήματα, υφίσταται σημαντική απόκλιση μεταξύ του συνολικού μαθηματικού σφάλματος και της αυστηρής μορφολογικής πιστότητας. Τα ευρήματα αυτά θέτουν τις βάσεις για μελλοντική έρευνα σε παραγωγικά μοντέλα (generative models) και μορφολογικά στοχευμένες συναρτήσεις απώλειας.

Acknowledgements

First and foremost, I would like to express my sincere gratitude to my supervisor, Professor Nikolaos Laskaris, for his invaluable guidance, continuous support, and the opportunity to work on such an innovative and challenging topic. His expertise in the field of Informatics and Signal Processing as well as his insightful suggestions were fundamental to the successful completion of this thesis.

I would also like to thank the Department of Informatics at the Aristotle University of Thessaloniki for providing the academic environment and resources necessary for my studies.

Furthermore, I am deeply grateful to my colleagues and friends for their encouragement and for the fruitful discussions we had throughout this journey. Their presence made the challenging moments of research and coding much more manageable.

Last but not least, I want to express my deepest appreciation to my family, my father Basileios, my mother Aikaterini, my older brother Giannis and my younger sister Eleni. This achievement would not have been possible without their unconditional love, patience, and support. I owe them everything for believing in me and standing by my side during all these years of my academic pursuit.

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Abbreviations

Acronym	What (it) Stands For
1D-CNN	1-Dimensional Convolutional Neural Network
3D	Three-Dimensional
ADM	Algebraic Distance Minimization
AF	Atrial Fibrillation
ANF	Adaptive Notch Filters
API	Application Programming Interface
AV	Atrioventricular
BiLSTM	Bidirectional Long Short-Term Memory
CBAM	Convolutional Block Attention Module
CPU	Central Processing Unit
CRBBB	Complete Right Bundle Branch Block
CTB	Channel-Time-Batch
CVD	Cardiovascular Disease
CW	Continuous Wave
DC	Direct Current / DC Offset
DWT	Discrete Wavelet Transform
ECC	Excitation-Contraction Coupling
ECG	Electrocardiogram
FMCW	Frequency Modulated Continuous Wave
GAN	Generative Adversarial Network
GAP	Global Average Pooling
GBD	Global Burden of Disease
GHz	Gigahertz
HRV	Heart Rate Variability

Abbreviations

I/Q	In-phase and Quadrature
IBI	Inter-Beat Interval
ICG	Impedance Cardiography
IF	Intermediate Frequency
IoT	Internet of Things
LFMCW	Linear Frequency Modulated Continuous Wave
MODWT	Maximal Overlap Discrete Wavelet Transform
MRA	Multiresolution Analysis
MSE	Mean Squared Error
NN	Neural Network
OOM	Out-Of-Memory
QRS	Q, R, and S waves (Ventricular depolarization complex)
RA	Receiving Antenna
RAM	Random Access Memory
RL	Reinforcement Learning
SB	Sinus Bradycardia
SE	Squeeze-and-Excitation
SNR	Signal-to-Noise Ratio
STach	Sinus Tachycardia
STFT	Short-Time Fourier Transform
TA	Transmitting Antenna
TFM	Task Force Monitor
VRAM	Video Random Access Memory

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Chapter 1

The Cardiovascular System and the ECG Signal

1.1 Human Heart Anatomy

The heart is the primary organ responsible for circulating blood throughout the body. Operating as a dual pump, it removes deoxygenated blood from the systemic circulation, pumps it to the lungs for oxygenation, and subsequently returns oxygenated blood to the rest of the body.

Anatomically, the human heart is divided into four distinct chambers: two superior chambers known as the left and right atria, and two inferior chambers called the left and right ventricles. The right ventricle's primary function is to pump oxygen-depleted blood into the lungs via the pulmonary trunk and arteries. Conversely, the left ventricle is responsible for forcefully ejecting oxygen-rich blood into the systemic circulation through the aorta [12].

1.2 The Cardiac Cycle

The cardiac cycle encompasses the physiological events that occur from the beginning of one heartbeat to the beginning of the next. It primarily consists of two alternating phases for both the atria and ventricles: systole (the contraction phase, during which blood is ejected) and diastole (the relaxation phase, during which the chambers fill with blood).

To ensure effective blood circulation, these mechanical events must be meticulously synchronized. In a healthy adult with a resting heart rate of 70–75 beats per minute, a single cardiac cycle lasts approximately 0.8 seconds [12].

1.2.1 Stages of the Cardiac Cycle

The cardiac cycle is characterized by significant shifts in chamber pressures and volumes. The main stages can be outlined as follows:

Atrial Systole and Diastole

Following electrical depolarization, the atria contract (atrial systole). This contraction increases pressure within the atria, forcing the remaining blood into the ventricles through the open atrioventricular (AV) valves (tricuspid and mitral valves). Atrial systole lasts for approximately 100 ms. Subsequently, the atrial muscles relax (atrial diastole) just before the onset of ventricular systole.

Ventricular Systole

As the ventricles begin to contract, the pressure inside the chambers rises rapidly, surpassing the pressure in the relaxed atria. This pressure gradient causes the AV valves to close, preventing backflow. Ventricular systole comprises three main sub-phases:

- **Isovolumetric Ventricular Contraction:** Muscle tension increases and ventricular pressure rises, but all valves (AV and semilunar) remain closed, resulting in no change in ventricular volume (lasting ~ 0.05 seconds).
- **Rapid Ventricular Ejection:** As ventricular pressure exceeds the pressure in the aorta and pulmonary arteries, the semilunar valves open. Approximately 70% of the ventricular blood is forcefully ejected (~ 0.13 seconds).
- **Reduced Ventricular Ejection:** The remaining 30% of the stroke volume is ejected as ventricular pressure begins to drop (~ 0.09 seconds).

Ventricular Diastole

Following repolarization, the ventricles relax. This diastolic phase lasts about 430 ms and is divided into:

- **Isovolumic Ventricular Relaxation:** Ventricular pressure drops, causing the semilunar valves to close (preventing backflow from the arteries). The AV valves are still closed, maintaining a constant blood volume.

- **Late Ventricular Diastole:** As pressure drops below the atrial pressure, the AV valves open. Blood flows passively from the major veins into the atria and directly into the ventricles, completing the cycle.

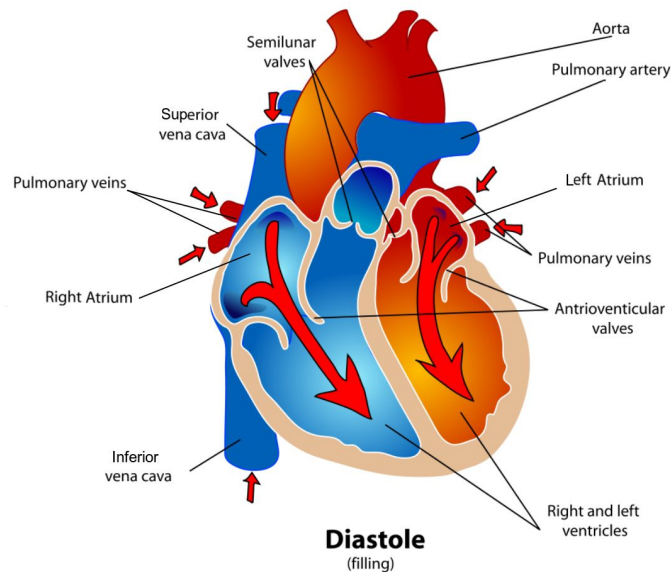


FIGURE 1.1: The relationship between the phases of the cardiac cycle (atrial and ventricular systole/diastole) and the corresponding waveforms (P, QRS, T) of the ECG signal.

1.2.2 Electromechanical Coupling and the ECG

The mechanical events of the cardiac cycle, governed by pressure gradients and fluid dynamics, are intricately synchronized with the heart's electrical activity. Fluids naturally flow from regions of higher pressure to lower pressure; however, in the heart, these pressure gradients are actively generated by muscle contractions triggered by electrical action potentials. This electrical activity can be captured and recorded non-invasively as an Electrocardiogram (ECG).

The distinct mechanical phases of the cardiac cycle directly correspond to the characteristic waveforms observed in an ECG signal:

- **The P wave:** This waveform represents the electrical depolarization of the atria, which is immediately followed by atrial systole. While the ventricles are approximately 70-80% filled passively during diastole, this atrial contraction (often referred to as the “atrial kick”) actively contributes the remaining 20-30% of the ventricular filling volume.
- **The QRS complex:** This prominent complex corresponds to the depolarization of the ventricles. This electrical signal triggers ventricular systole. The rapid increase in pressure closes the atrioventricular valves (isovolumic contraction) and subsequently forces

the semilunar valves open to eject blood (stroke volume) into the aorta and pulmonary trunk.

- **The T wave:** This waveform signifies ventricular repolarization, which precedes ventricular diastole (relaxation). As the ventricular muscle relaxes, the internal pressure drops, causing the semilunar valves to close and allowing the chambers to eventually refill with blood for the next cycle.

Understanding this elegant electromechanical synchronization is crucial, as any physiological abnormality or external interference—such as respiratory movement—will manifest as variations or noise within the extracted ECG signal.

1.3 The ECG Signal and the Respiration Artifact

Clinical Significance of ECG Morphology

The clinical utility of any electrocardiographic (ECG) monitoring system is fundamentally rooted in its ability to accurately capture not only the general heart rate but also specific, subtle waveform characteristics that indicate underlying pathologies. In the context of contactless monitoring, preserving the precise morphological integrity of the reconstructed ECG signal is of paramount importance. Accurate detection of basic rhythm-related abnormalities, such as Sinus Tachycardia (STach) and Sinus Bradycardia (SB), requires highly precise Inter-Beat Interval (IBI) measurement to differentiate normal physiological variations from conditions like congestive heart failure or acute myocardial infarction.[8].

Furthermore, capturing structural anomalies and complex arrhythmias demands immense signal clarity. For instance, Atrial Fibrillation (AF)—a widespread and clinically significant arrhythmia—is characterized by irregular QRS complexes, the absence of discernible P-waves, and an undulating baseline caused by chaotic fibrillatory waves (f-waves). Similarly, diagnosing waveform-related abnormalities, such as ST-segment depression (a hallmark of myocardial ischemia) and T-wave alterations, demands millimeter-level precision in amplitude reconstruction. Conduction disorders, like Complete Right Bundle Branch Block (CRBBB), also rely on the clear depiction of widened QRS complexes (≥ 120 ms) and specific morphological patterns, such as the secondary R' wave.[8].

The Respiration Artifact Challenge

To capture these vital diagnostic features unobtrusively, radar sensors continuously monitor the mechanical vibrations of the chest wall. However, the raw radar signal is inherently a

composite of two primary physiological activities: the heartbeat and respiration. While the cardiac micro-motions contain the essential mechanical signatures required to synthesize the electrical ECG waveforms, the physical displacement of the chest caused by breathing is significantly larger in amplitude. Consequently, respiration acts as a massive low-frequency artifact that dominates the acquired data. Beyond respiration, reliable extraction of the cardiac signal faces significant challenges because the mechanical vibration caused solely by the heartbeat is inherently extremely weak. This results in a notoriously low Signal-to-Noise Ratio (SNR). Furthermore, the acquired signal is often corrupted by random bodily movements (motion artifacts), inherent system noise, and extraneous reflections from people or objects in the patient's immediate vicinity. If this complex mixture of respiratory and environmental noise is not robustly filtered, its inherent baseline wander can easily mask the crucial diagnostic markers mentioned above—such as f-waves or ST-segment deviations—rendering the system ineffective for clinical-grade diagnostics.[2]

Spatial Filtering and Phase Unwrapping

Therefore, accurately extracting the cardiac signal requires overcoming this low SNR and mitigating this overwhelming respiratory artifact. Recent advancements in radar signal processing and innovative hardware design leverage sophisticated spatial filtering techniques, such as 3D beamforming, to address this challenge. By applying 3D beamforming and specialized antenna designs, the system can focus the electromagnetic waves on a specific, narrow strip along the patient's chest, spatially isolating the specific cardiac motion from the broader respiratory movements and environmental clutter. Mathematically, the raw radar data are reconstructed into spatial reflections across azimuth and elevation using the 3D beamforming formulation:

$$\mathbf{S}_t(r, \theta, \phi) = \mathbf{a}_t(r, \theta, \phi) \exp(j\omega_t(r, \theta, \phi))$$

where $\mathbf{a}_t$ and ω_t encode the amplitude and phase information, respectively, both of which are heavily modulated by chest movements [8].

Once the optimal spatial point is targeted, the system processes the raw intermediate frequency data, typically represented as In-phase (I) and Quadrature (Q) components. Meticulous filtering of these I/Q components and the extraction of the precise phase signal heavily suppresses the respiration artifact. Specifically, the chest displacement $r(t)$ is directly encoded into the phase of the intermediate frequency (IF) signal, mathematically defined as:

$$\phi_{\text{IF}}(t) = 4\pi \frac{r(t) + d}{\lambda}$$

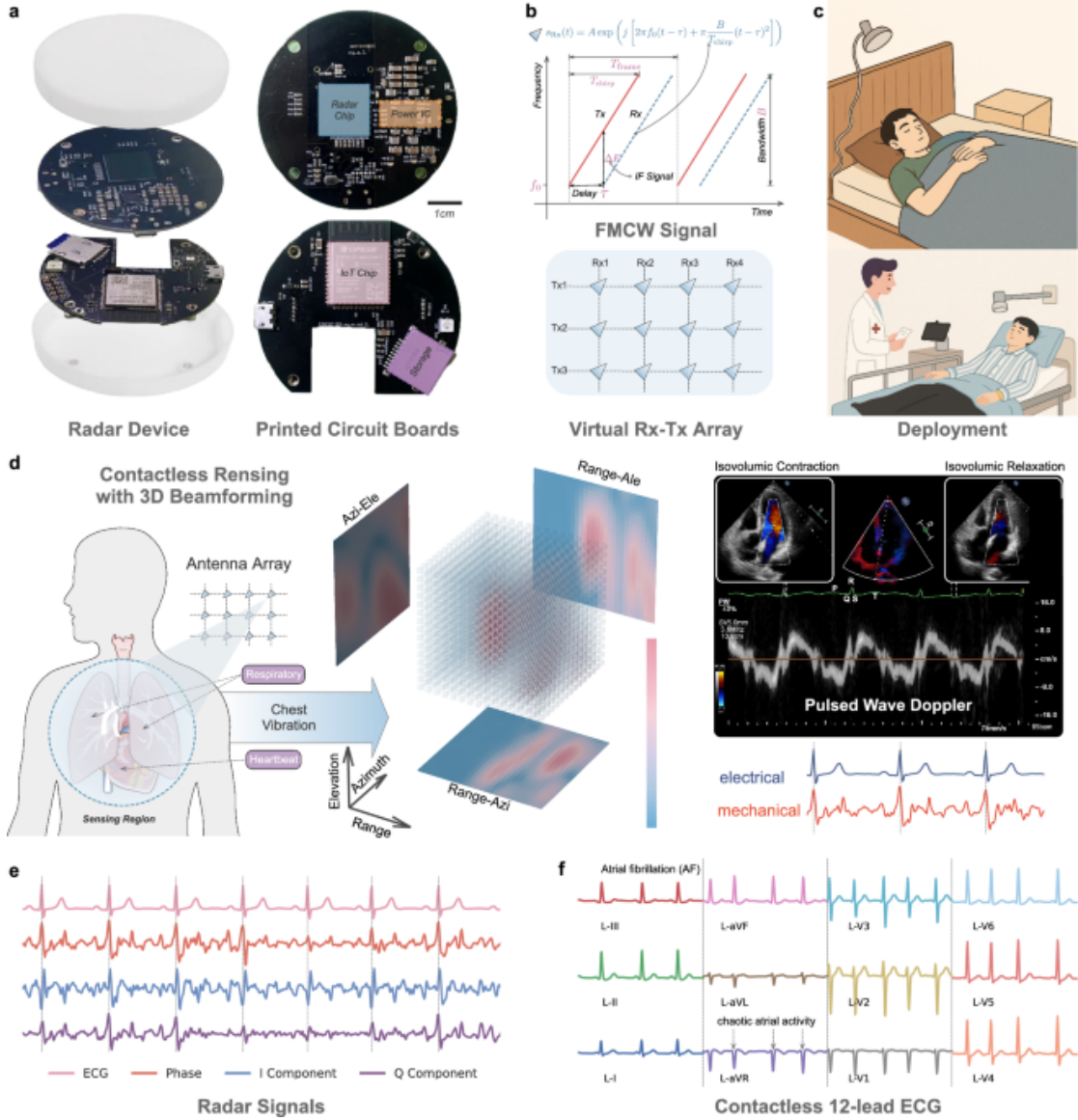


FIGURE 1.2: Overview of contactless cardiac sensing. The system utilizes 3D beamforming to spatially isolate chest vibrations, subsequently unwrapping the phase to extract mechanical cardiac signals hidden within respiratory artifacts.

where d is the subject's distance and λ is the radar wavelength. Because the amplitude of respiratory motion is vastly larger than cardiac motion, subtle heartbeat movements must be extracted by unwrapping this phase. To further suppress the low-frequency respiratory baseline and amplify the subtle high-frequency cardiac features (such as R-peaks), a noise-robust differentiator is often applied [8]. For instance, a common discrete differentiation filter can be expressed as:

$$f'' = \frac{(f_3 + f_{-3}) + 2(f_2 + f_{-2}) - (f_1 + f_{-1}) - 4f_0}{16\Delta^2}$$

where f_i represents the unwrapped phase sample at time index i , and Δ is the sampling interval. This mathematical processing significantly improves the visibility of critical cardiac wave peaks even under severe respiratory noise. Ultimately, this successful mitigation yields a clean

mechanical waveform of the heartbeat, serving as the fundamental input for the deep learning models responsible for reconstructing a clinically valid 12-lead ECG.

1.3.1 The Challenge of Respiratory Harmonics and Static Clutter

While standard filtering techniques can address basic baseline wander, non-contact Continuous Wave (CW) and Frequency-Modulated Continuous Wave (FMCW) radar systems face significantly more complex challenges when isolating the heartbeat from respiration. The mechanical displacement of the chest wall caused by breathing is vastly larger in amplitude than the displacement caused by the heartbeat, effectively masking the cardiac signal.

Recent literature highlights two primary sources of interference in radar-based vital sign monitoring:

- **Static Clutter and DC Offset:** Reflections from stationary objects in the environment, as well as internal radar system noise, create low-frequency interference (static noise). This often manifests as a center offset in the complex plane of the radar's In-phase and Quadrature (I/Q) signals.
- **Respiratory Harmonics:** The fundamental frequency of human respiration is typically between 0.2 and 0.4 Hz. However, the respiration signal is not a perfect sine wave. Consequently, it generates higher-order harmonics (especially the 2nd, 3rd, and 4th harmonics). Because the normal resting heart rate lies between 1.0 and 1.5 Hz (60-90 beats per minute), these respiratory harmonics fall directly into the frequency band of the heartbeat, making simple band-pass filtering ineffective [7].

To overcome these obstacles, state-of-the-art approaches propose sophisticated signal processing at the software level. For instance, recent studies suggest the use of Algebraic Distance Minimization (ADM) techniques to accurately estimate and calibrate the center offset of the I/Q signals in the complex plane, thereby suppressing static clutter. Furthermore, once the baseline is corrected, Adaptive Notch Filters (ANF) can be dynamically deployed to track and explicitly eliminate the interfering respiratory harmonics without destroying the underlying cardiac signal [7].

Such methodologies emphasize the need for advanced mathematical transformations and adaptive filtering—or alternatively, robust Deep Learning architectures—to extract a clean ECG-equivalent signal from raw radar data.

Chapter 2

Non-Contact Radar Monitoring

2.1 Continuous Wave Radars and Unobtrusive Health Monitoring

The Need for Contactless Monitoring

Cardiovascular diseases (CVDs) remain the leading cause of mortality worldwide, imposing immense pressure on global healthcare systems. According to the Global Burden of Disease (GBD) estimates, the number of deaths from CVDs is projected to reach 20.5 million by 2025 and surge to 35.6 million by 2050. The progression of these diseases often exhibits insidious and paroxysmal characteristics, highlighting a pressing need for dynamic, out-of-hospital continuous monitoring. The traditional Electrocardiogram (ECG) remains the clinical gold standard; however, it requires the direct attachment of adhesive electrodes. For prolonged, routine monitoring—especially during sleep where frequent postural changes occur—this contact-based approach can induce skin irritation, restrict daily activities, and lead to electrode detachment. Consequently, there is a critical demand for alternative, contactless monitoring solutions.[3]

Principles of Radar-Based Sensing

To address these clinical challenges, radar-based sensor systems—specifically utilizing CW radars—have emerged as a revolutionary technology for unobtrusive health monitoring. These systems operate by emitting high-frequency electromagnetic waves that propagate toward the patient and reflect off their chest. The radar system detects the phase shifts and micro-movements on the body surface that are mechanically generated by cardiac contractions and respiratory activity. Unlike conventional ECGs, this non-invasive approach requires absolutely no physical contact. Furthermore, the electromagnetic waves can penetrate non-conductive materials, allowing the sensor’s compact housing to be discreetly placed under a hospital bed

to record vital signs accurately through mattresses and blankets.[5].

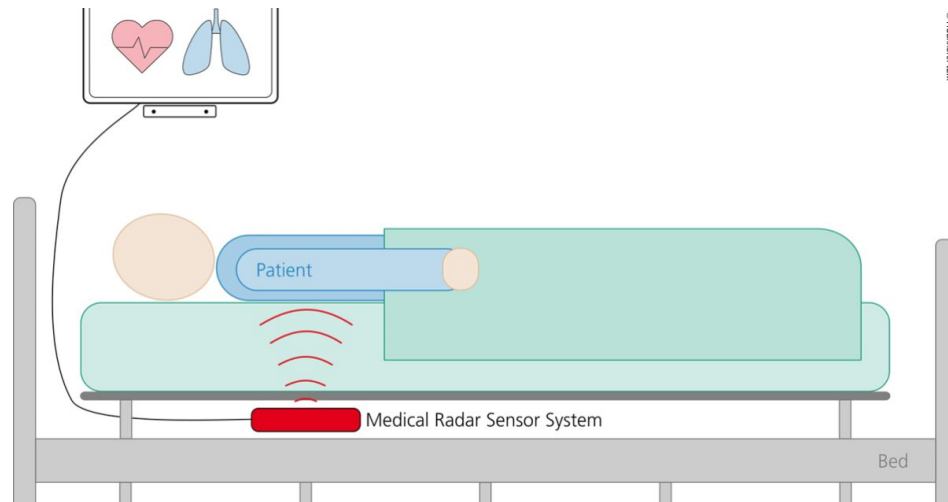


FIGURE 2.1: Conceptual illustration of an unobtrusive radar-based health monitoring system positioned discreetly beneath a hospital bed.

Clinical Implementation and Advantages

The integration of CW radar technology into clinical settings significantly enhances patient comfort while simultaneously alleviating the heavy workload imposed on healthcare personnel. By eliminating routine manual tasks, such as applying, maintaining, or repositioning electrodes, medical staff can allocate their time more efficiently. Moreover, continuous and unobtrusive monitoring facilitates the early detection of cardiovascular and respiratory abnormalities, thereby increasing the likelihood of timely medical intervention. The technology is also designed for maximum flexibility; future iterations aim to be mounted on walls or ceilings and upgraded to multichannel solutions capable of monitoring multiple patients simultaneously.

While the concept of wireless, bedside monitoring is not entirely novel, radar technology represents a significant leap forward in the field. Previous commercial solutions have largely relied on sensor mats, such as the EarlySense or SafeSense3 systems, which are placed under the patient to detect movement, respiratory rates, and heart rates via changes in air pressure. Although these systems are effective in assisting nurses by providing basic vital signs and bed-exit alarms, radar-based monitoring offers a superior level of true, contactless precision. By remotely capturing complex cardiopulmonary signals, radar solutions offer a future-proof, patient-centric approach to continuous monitoring, with broad applications extending from intensive care units and nursing homes to vehicular environments.[5].

Furthermore, while wearable devices such as wrist-worn sensors and chest patches have gained popularity for long-term monitoring, they are inherently limited. Wearables are often constrained by limited-lead configurations, providing a restricted view of cardiac activity compared to a full 12-lead assessment. [8].

A major advantage of radar-based ECG monitoring lies in its cost-effectiveness, which facilitates large-scale clinical deployment. Traditional 12-lead ECG machines are high-cost medical instruments, often exceeding \$1,000 per unit. In contrast, modern radar solutions can be developed using off-the-shelf components, with core radar chips costing as little as \$20 and total system costs remaining under \$30. This drastic reduction in cost enables the implementation of high-quality cardiac monitoring in resource-limited environments and home-care settings, where cardiovascular disease (CVD) imposes a significant economic burden.[8].

Beyond its low-cost architecture, the translational potential of this technology relies heavily on modern Internet of Things (IoT) integration. The device’s wireless communication modules allow for the real-time transmission of raw radar data to server-based reconstruction backends. From there, the reconstructed ECG signals can be seamlessly forwarded to hospital telemetry systems or home-based IoT hubs via lightweight gateways, outlining a highly practical route for integrating contactless monitoring into existing healthcare infrastructures [8].

Advanced FMCW Technology

While basic CW radar is effective for motion detection, recent advancements have integrated Linear Frequency Modulated Continuous Wave (LFMCW) technology, typically operating at higher frequencies such as 60 GHz. In such systems, the transmitted chirps follow a linearly increasing frequency over time, which can be mathematically expressed as:

$$f(t) = f_0 + \frac{B}{T_{\text{chirp}}}t$$

where f_0 is the initial frequency, B denotes the bandwidth, and T_{chirp} is the chirp duration. This higher bandwidth, combined with 3D beamforming techniques, allows for superior spatial resolution. This precision is what enables the system to isolate subtle chest vibrations and reconstruct complex electrical waveforms, transforming mechanical micro-motions into a clinically valid 12-lead ECG signal.

The viability of this advanced FMCW technology is no longer purely theoretical; recent large-scale clinical studies have validated radar-based ECG reconstruction in cohorts of nearly 7,000

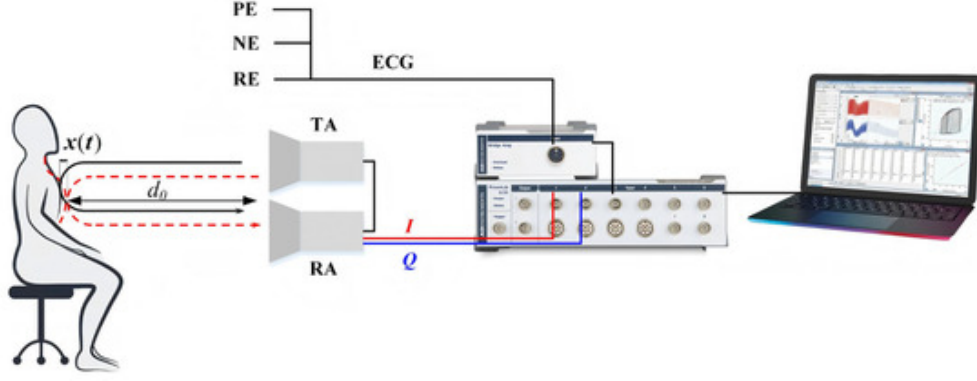


FIGURE 2.2: Schematic representation of a synchronized data acquisition setup, capturing continuous-wave radar echoes (In-phase and Quadrature channels) alongside standard ECG signals.

individuals. These systems have demonstrated high correlation (median 0.77) with ground-truth ECG and low error rates in inter-beat interval (IBI) detection, proving their reliability for continuous, overnight, and long-term cardiovascular care.[8].

To ensure the reliability of long-term and overnight continuous monitoring, advanced radar systems incorporate automated artifact exclusion procedures. For instance, during sleep monitoring, periods of large, random bodily movements can distort the signal. By establishing a specific amplitude threshold (e.g., excluding segments where the predicted ECG amplitude remains continuously below 0.2 mV), the system can automatically filter out these motion-related artifacts. This method successfully retains highly reliable data, demonstrating strong correlation with ground-truth Holter ECGs and maintaining stable Inter-Beat Interval (IBI) tracking throughout the night [8].

2.1.1 Mathematical Modeling of the Radar Signal and I/Q Demodulation

In Frequency-Modulated Continuous Wave (FMCW) and Continuous Wave (CW) radar systems used for vital sign monitoring, the transmitted electromagnetic wave reflects off the human chest wall. The physiological movements—specifically, the macroscopic chest displacement due to respiration and the microscopic vibrations from the heartbeat—modulate the phase of the reflected radar signal.

When the received echo is mixed with the transmitted signal, an Intermediate Frequency (IF) signal is produced. This IF signal is typically processed in the complex plane, yielding an In-phase ($I(t)$) and a Quadrature ($Q(t)$) component. The phase variations $\phi(t)$ of the IF signal directly correlate with the chest wall displacement, encompassing both respiratory and cardiac activity [7].

Ideally, plotting the $I(t)$ and $Q(t)$ signals in the complex plane traces a perfect arc centered at the origin. The physiological phase $\phi(t)$ can then be straightforwardly extracted using the four-quadrant inverse tangent function:

$$\phi(t) = \text{atan2}(Q(t), I(t)) \quad (2.1)$$

However, in real-world scenarios, reflections from stationary objects in the background (static clutter) and internal hardware imperfections introduce significant DC offsets (dc_I and dc_Q). Consequently, the center of the arc shifts away from the origin, heavily distorting the extracted phase and burying the subtle cardiac signal beneath low-frequency noise.

To mathematically resolve this, recent literature emphasizes the application of Algebraic Distance Minimization (ADM). ADM is an advanced fitting technique that accurately estimates the true center of the distorted I/Q trajectory in the complex plane. By actively calibrating this center offset, ADM suppresses the static environmental noise and reconstructs a signal with a high Signal-to-Noise Ratio (SNR). This rigorous preprocessing is a vital step before any advanced filtering or Deep Learning models can be reliably applied to isolate the heartbeat from the respiratory harmonics [7].

Chapter 3

Signal Processing and Deep Learning

3.1 Deep Learning for Time-Series Analysis

Data-Driven Inverse Modeling

While robust spatial filtering isolates the cardiac mechanical waveform, translating this physical displacement into a detailed electrical signal remains a highly complex challenge. Early research attempted to extract electrophysiological features from mechanical motions using traditional, linear signal processing methods. However, the physiological mapping dictated by the ECC mechanism is fundamentally non-linear. Consequently, traditional mathematical transformations struggle to model the intricate, patient-specific mechano-electrical coupling of the heart. To overcome this, data-driven inverse modeling using Deep Learning has emerged as the state-of-the-art methodology, specifically designed to approximate these complex non-linear relationships and synthesize continuous ECGs from sequential time-series data.[3]

Feature Extraction with 1D-CNNs

To effectively decode this mechanical time-series data, modern architectures typically employ 1-Dimensional Convolutional Neural Networks (1D-CNNs) as the primary feature extractors. Unlike their 2D counterparts predominantly used in image processing, 1D-CNNs are specifically designed to slide one-dimensional convolutional filters across sequential signals. This operation allows the network to automatically detect and extract hierarchical local features without relying on manual feature engineering. In the context of radar ECG, 1D-CNNs excel

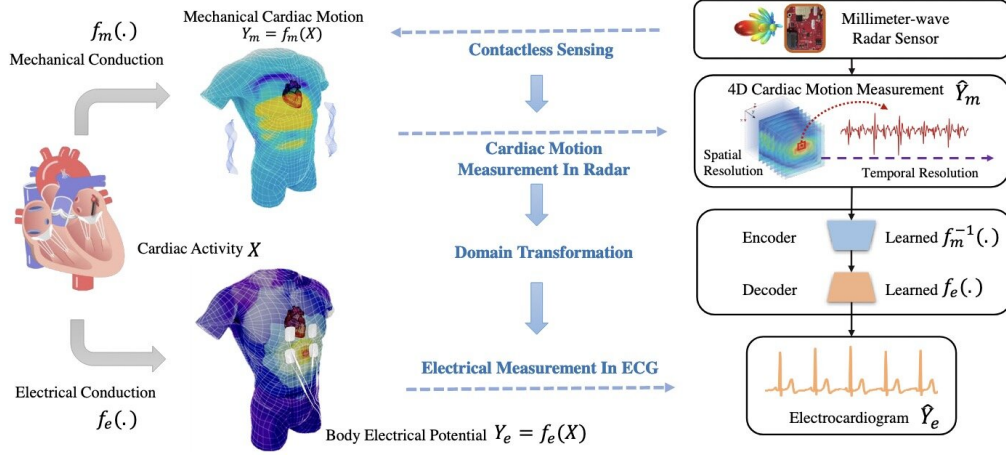


FIGURE 3.1: The theoretical framework of data-driven inverse modeling based on the Excitation-Contraction Coupling (ECC) mechanism. The deep learning network learns the complex non-linear mapping between mechanical cardiac motion (captured via radar) and electrical cardiac activity (ECG).

at identifying the sharp peaks, subtle inflections, and specific morphological patterns corresponding to vital electrical events, such as the QRS complex or T-waves, hidden within the raw mechanical vibrations [6].

Temporal Modeling and Network Architecture

However, extracting local morphological features is insufficient for full ECG reconstruction, as cardiac rhythms are dictated by long-term temporal dependencies. To address this, the local features extracted by the 1D-CNN are subsequently processed by Bidirectional Long Short-Term Memory (BiLSTM) networks. While standard LSTMs successfully mitigate the vanishing gradient problem to retain historical context in long sequences, BiLSTMs advance this by processing the data in both forward and backward directions [14]. This bidirectional architecture ensures that the prediction of a specific electrical waveform at any given time step is informed by both preceding and succeeding mechanical events. This dual-context approach is critical for accurately capturing Inter-Beat Intervals (IBI) and modeling the overall physiological rhythm of the heart.

Beyond core feature extraction and temporal modeling, state-of-the-art reconstruction models often operate on an encoder-decoder principle. The initial 1D-CNN layers, coupled with Max Pooling operations, act as an encoder that downsamples and compresses the radar signal into a lower-dimensional latent representation, effectively discarding redundant noise. To reconstruct this compressed representation back into the high-resolution time-domain of an ECG, Transposed 1-Dimensional Convolutions (Transposed 1D-CNNs) are utilized. These specialized layers upsample the feature maps, learning to expand the latent features back to the

required sequence length before the data is fed into the BiLSTM and subsequent Fully Connected (Dense) layers for the final value regression.

Regularization and Optimization

Furthermore, training deep networks on highly variable physiological data introduces a significant risk of overfitting. To ensure the model generalizes well to unseen patients and varying clinical conditions, robust regularization techniques are integrated within the convolutional blocks. Batch Normalization is employed to stabilize and accelerate the learning process by standardizing the inputs to a layer, while Dropout layers randomly deactivate a fraction of neurons during training, forcing the network to learn robust, redundant features rather than memorizing the training set. Ultimately, the network optimizes its trainable parameters by minimizing the Mean Squared Error (MSE) between the predicted output and the ground-truth ECG, ensuring a mathematically rigorous convergence towards the true electrical waveform.

Ultimately, the synergy of these two neural network architectures creates a robust and highly accurate computational pipeline. The 1D-CNN isolates the precise spatial morphology of the individual cardiac waveforms, while the BiLSTM establishes the contextual rhythm and temporal dynamics. Together, they bridge the complex gap between mechanical chest vibrations and electrical activity, enabling the true, contactless reconstruction of a diagnostic-grade 12-lead ECG [8].

3.2 Wavelet Transforms and Multiresolution Analysis

Limitations of Traditional Transforms

Biological signals, such as the cardiopulmonary micro-motions captured by continuous wave radars, are inherently non-stationary. This means their frequency components and statistical properties change continuously over time. Traditional signal processing methods, most notably the standard Fourier Transform, are highly effective at identifying the frequency content of a signal but completely lose all temporal information. In other words, Fourier analysis can reveal which frequencies are present, but not when they occur. While the Short-Time Fourier Transform (STFT) attempts to address this by windowing the signal into smaller segments, it suffers from a fixed resolution trade-off dictated by the Heisenberg-Gabor uncertainty principle: a narrow time window provides good time resolution but poor frequency resolution, and vice versa.

Multiresolution Analysis (MRA)

To overcome this critical limitation in time-series analysis, Wavelet Transforms are utilized. By employing a scalable mathematical function known as a "wavelet," this transform provides a dynamic, multi-scale time-frequency representation. It uses short time windows to precisely localize high-frequency events (such as the sharp R-peaks of an ECG or motion artifacts) and long time windows to capture low-frequency trends (such as the underlying respiratory rhythm), making it an optimal mathematical tool for analyzing non-stationary physiological signals [9].

To effectively isolate the cardiac signature from respiratory artifacts and background noise, the Wavelet Transform is typically applied through Multiresolution Analysis (MRA). Mathematically, MRA is implemented using a digital cascaded filter bank driven by two complementary functions: the scaling function (often referred to as the "father wavelet"), which acts as a low-pass filter, and the wavelet function (the "mother wavelet"), which acts as a high-pass filter. As the raw radar time-series passes through this filter bank, it is systematically decomposed into hierarchical levels of resolution. At each level, the high-pass filter extracts the "detail" coefficients (representing fast changes), while the low-pass filter extracts the "approximation" coefficients (representing slow, baseline trends). By selectively retaining specific detail levels and discarding the approximations that correspond to the low-frequency respiratory wander, MRA can mathematically purify the signal before it reaches the predictive models [9].

Integration with Deep Learning (MODWT)

However, integrating wavelet-based filtering directly into a Deep Learning pipeline introduces a significant structural challenge. The classical Discrete Wavelet Transform (DWT) heavily relies on decimation, or downsampling by a factor of two, at each decomposition level. While computationally efficient, this downsampling renders the DWT shift-variant, meaning that a slight time shift in the input signal drastically alters the resulting wavelet coefficients. This is highly detrimental for time-series forecasting and sequence-to-sequence neural networks, which require consistent temporal alignment.

To solve this, the Maximal Overlap Discrete Wavelet Transform (MODWT) is employed. Unlike DWT, MODWT eliminates the downsampling step entirely, maintaining the full length of the original sequence at every decomposition level. While this introduces redundant information compared to classical transforms, this redundancy is highly beneficial for deep learning models. It ensures circular shift-invariance, meaning the temporal alignment of the extracted features exactly matches the original radar time-series. This crucial modification provides the subsequent 1D-Convolutional and BiLSTM layers with highly stable, time-aligned frequency features without losing any temporal resolution [11].

Finally, the effectiveness of the MODWT heavily depends on the selection of the appropriate mother wavelet. For electrocardiographic reconstruction, it is imperative to use a wavelet that minimizes phase distortion, thereby preserving the exact morphology and alignment of the physiological waveform. The Symlet family, specifically the "sym4" wavelet, is widely adopted for this purpose. Symlets are designed to be nearly symmetrical, ensuring near-linear phase characteristics. When integrated as a feature extraction layer within a deep learning architecture, the combination of MODWT and the sym4 wavelet decomposes the complex, noisy radar signal into highly informative and perfectly synchronized multiresolution features, ultimately enhancing the network's reconstruction accuracy [10].

3.2.1 Motivation for Wavelet-Based Deep Learning Frameworks

Traditional and adaptive signal processing techniques, such as the Adaptive Notch Filters (ANF) combined with Algebraic Distance Minimization (ADM) proposed in contemporary studies [7], demonstrate remarkable efficacy in tracking and eliminating specific respiratory harmonics under stable conditions. However, radar-reflected vital sign signals are inherently non-stationary and highly prone to sudden non-linear distortions caused by minor random body movements or variations in breathing patterns.

While adaptive filters perform well in the frequency domain, they often introduce latency or struggle when the respiratory and cardiac frequencies overlap dynamically. To mitigate these limitations, this thesis leverages the Multiresolution Analysis (MRA) of Wavelet Transforms. Wavelets provide simultaneous time and frequency localization, making them uniquely suited for capturing transient features and isolating non-stationary noise. By feeding these multi-scale wavelet representations into deep 1D Convolutional Neural Networks (1D-CNNs), the system can automatically learn complex, non-linear mappings to reconstruct the target ECG signal, offering a robust alternative to purely analytical or adaptive filtering frameworks.

Chapter 4

Methodology and Implementation

4.1 Existing Frameworks & Code Origin

The foundational signal processing pipeline and deep learning architectures examined in this preliminary stage are based on the official MathWorks documentation for human health monitoring via continuous wave radar. It is important to note that while earlier iterations of this codebase required significant technical interventions to resolve API incompatibilities and dimension mismatches, the official implementation now executes seamlessly within the modern MATLAB R2026a environment. This out-of-the-box stability provides a robust and reliable baseline framework for further experimental investigation [10].

The primary objective of this reference framework is to address the overwhelming respiration artifact inherent in contactless radar data, where large-scale chest displacements effectively mask the subtle mechanical micro-motions of the heart. To train and evaluate its architectures, the MathWorks implementation utilizes a heavily preprocessed, machine-learning-ready subset of a clinical dataset originally published by Schellenberger et al. (2020). Specifically, the raw continuous-wave radar data was externally pre-demodulated to extract the phase, down-sampled to a stable frequency of 200 Hz, and strictly segmented into 1024-sample windows (representing approximately 5.12 seconds of recording) alongside synchronized ECG ground truth. Based on this rigidly formatted input, the framework proposes and compares two distinct neural network architectures: a baseline temporal model (Net 1) and a wavelet-enhanced model (Net 2) [7, 8, 10].

Baseline Architecture (Net 1)

The baseline model acts as a standard control, attempting to reconstruct the electrical ECG signal directly from the raw radar time-series. It employs a hybrid 1D-Convolutional Autoencoder combined with a Bidirectional Long Short-Term Memory (BiLSTM) network. Specifically, the raw 1024-point sequence is aggressively downsampled via primary encoder layers utilizing 64 filters with a large stride of 8, effectively discarding high-frequency noise. The data is then compressed into a latent space via Max Pooling, and subsequently upsampled through transposed convolutions (32 and 64 filters). Finally, the BiLSTM analyzes the sequence bidirectionally (using 8 hidden units) to model the temporal rhythm, before passing the data to fully connected layers for final value regression. The model utilizes the Adam optimizer with an initial learning rate of 0.001 over 800 epochs [6, 10, 14].

However, as demonstrated in the framework’s baseline evaluation, this direct approach fails to accurately capture the cardiac signature. The network becomes "blinded" by the massive low-frequency respiratory baseline wander, resulting in morphologically distorted waveforms, misaligned R-peaks, and widespread reconstruction errors [7, 8].

Wavelet-Enhanced Architecture (Net 2) & Multiresolution Analysis

To overcome the limitations of the baseline model, the core contribution of the reference MathWorks implementation is the integration of a Multiresolution Analysis (MRA) framework directly into the deep learning pipeline. This forms the proposed Wavelet-Enhanced Architecture (Net 2) [9, 10].

The fundamental difference lies in the feature extraction phase. Before entering the convolutional blocks, the raw signal passes through a custom `modwtLayer`. Utilizing the Maximal Overlap Discrete Wavelet Transform (MODWT) with a Symlet-4 (`sym4`) mother wavelet, the signal is decomposed into hierarchical frequency levels [10, 11].

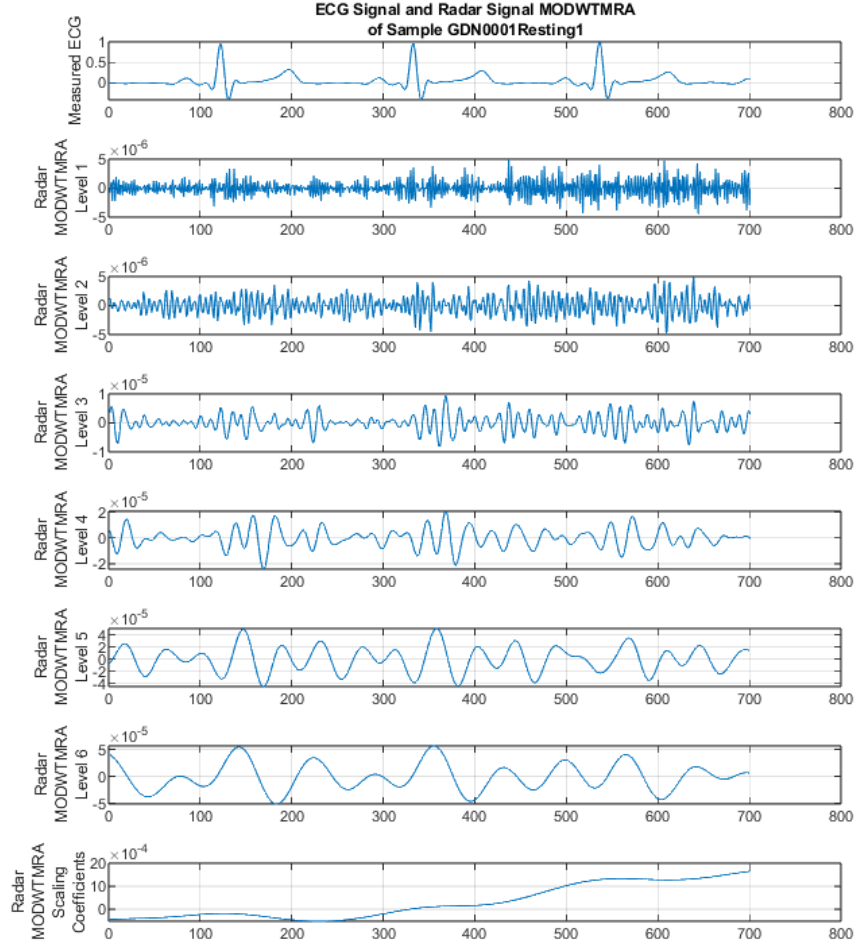


FIGURE 4.1: Multiresolution Analysis (MODWTMRA) of a radar sample. The intermediate detail levels clearly isolate the mechanical cardiac cycles, while discarding the low-frequency respiratory baseline.

Crucially, the network is configured to mathematically discard the scaling coefficients (which contain the dominant respiratory artifacts) and strictly preserve only the intermediate detail levels (Levels 3 through 5) where the mechanical cardiac energy is concentrated [11].

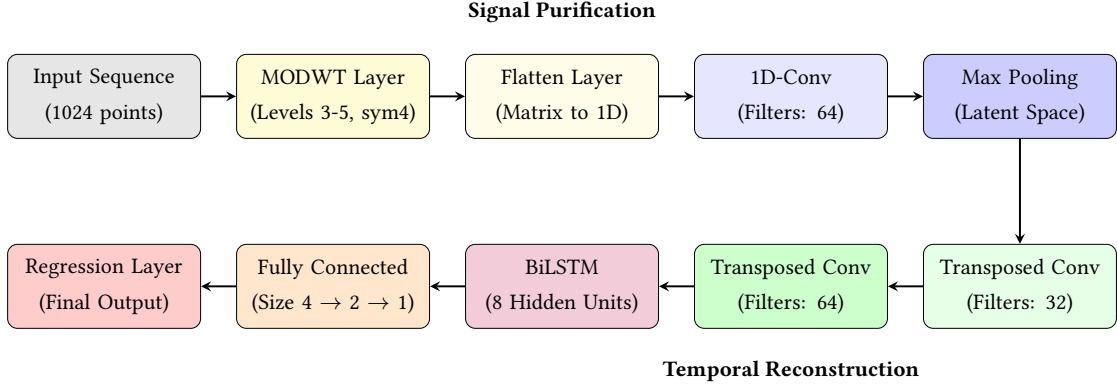


FIGURE 4.2: Block diagram of the Wavelet-Enhanced Architecture (Net 2). The raw signal is strictly decomposed via MODWT to isolate levels 3 through 5, mathematically flattened, and then processed through the standard convolutional and BiLSTM blocks.

By mathematically purifying the sequence and collapsing it back into a 1D format (via a flatten-Layer) before it enters the standard 1D-CNN and BiLSTM blocks, Net 2 successfully maps the complex morphology of the heartbeat. From a technical standpoint, integrating this specialized wavelet layer into traditional training pipelines historically required appending a `strictRegressionLayer()` at the terminal node to ensure proper `DAGNetwork` compilation. This structural adjustment, coupled with parallel CPU multi-threading, successfully optimizes computational workloads, reducing training times by up to 60% [6, 10].

Performance Evaluation of the Reference Framework

To substantiate the architectural superiority of the wavelet-enhanced model, the reference framework evaluates both networks using a synchronized loss metric. Because the modern APIs utilized by Net 1 compute the Sum of Squared Errors while Net 2 computes the Mean Squared Error, the validation metrics are mathematically scaled by a factor of 1024 for direct comparison [10].

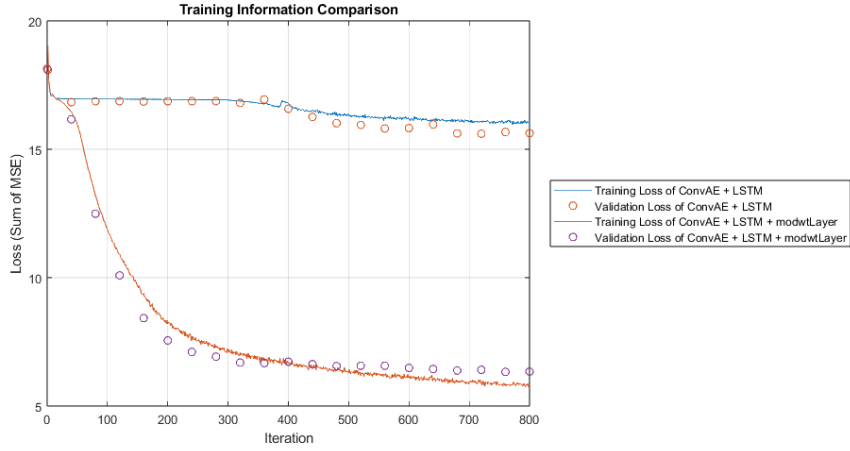


FIGURE 4.3: Comparative Training and Validation Loss history. The wavelet-enhanced model (Net 2) clearly achieves superior and significantly faster convergence compared to the baseline temporal model (Net 1).

As illustrated in Figure 4.3, Net 1 struggles with convergence, plateauing at a high error rate of approximately 16.03. Conversely, Net 2 not only converges significantly faster but also reaches a drastically lower and highly stable error floor of roughly 5.84 (scaled). When applied to unseen test data, customized inference pipelines demonstrate that the test MSEs for Net 2 exhibit a stark leftward shift, heavily concentrating reconstruction errors near zero. This statistical superiority directly translates to accurate morphological alignment and precise R-peak detection in the reconstructed ECG waveforms. Consequently, this proven wavelet-augmented architecture (Net 2) serves as the theoretical and structural foundation for the subsequent customized experiments conducted in this thesis [8, 10].

4.2 Original Dataset Acquisition

The empirical foundation for this research is based on a comprehensive clinical dataset initially introduced by Schellenberger et al. (2020) and published in *Scientific Data*. Unlike heavily pre-processed subsets, this original repository contains approximately 24 hours of raw, continuous-wave radar echoes and synchronized electrical reference signals, recorded from 30 healthy volunteers (14 male, 16 female) in a controlled clinical environment. The dataset is publicly distributed under the Creative Commons Attribution 4.0 International Public License (CC BY 4.0), which permits the unrestricted adaptation and modification of the raw data for advanced research [13].

The data acquisition architecture was designed to capture high-fidelity physiological micro-motions. The primary sensor was a Six-Port continuous-wave radar system operating at 24 GHz, which sampled the raw In-Phase (I) and Quadrature (Q) signals at a high frequency of

2000 Hz. Simultaneously, a Task Force Monitor (TFM 3040i) was utilized as the clinical reference device, recording a 3-lead ECG at 1000 Hz, impedance cardiography (ICG) at 500 Hz, and continuous blood pressure at 100 Hz. This synchronized setup ensures that every mechanical displacement captured by the radar can be perfectly mapped to a medical-grade electrical ground truth [13].

To ensure that any developed signal processing algorithms can generalize across highly dynamic and challenging physiological states, the clinical protocol recorded the subjects under five distinct scenarios, each designed to trigger specific hemodynamic and autonomic nervous system responses:

- **Resting (Spontaneous Breathing):** The standard baseline state lasting at least 10 minutes, where subjects lay relaxed with normal breathing. Here, the dominant respiratory motion heavily overlaps with the mechanical cardiac signature.
- **Valsalva Maneuver:** Subjects forcefully exhaled against a closed glottis for 20 seconds. This induces rapid transient fluctuations in blood pressure and Heart Rate Variability (HRV), testing a model’s ability to track dynamic changes.
- **Apnea (Breath-Holding):** Subjects completely ceased breathing after either a full inhalation or exhalation. This completely eliminates the respiratory artifact, yielding a purely isolated mechanical cardiac signal for algorithmic validation.
- **Tilt Up:** The tilt table was raised to 70 degrees, provoking a strong reaction from the autonomic nervous system and significantly altering blood pressure and heart rate.
- **Tilt Down:** The table was returned to the 0-degree baseline, with continued recording to monitor the physiological recovery.

Data Synchronization and Raw Signal Characteristics

A critical requirement for training supervised deep learning models on time-series data is the absolute temporal alignment between the input features and the ground-truth labels. Because the radar system and the Task Force Monitor operated on independent hardware clocks with different sampling frequencies (2000 Hz and 1000 Hz, respectively), an explicit synchronization mechanism was required. To achieve this, the original researchers utilized a Gold code binary sequence generated by a microcontroller. This sequence was sampled simultaneously by both the radar’s analog-to-digital converter and the external input of the TFM. By mathematically applying cross-correlation to these recorded sequences during post-processing, the exact time lag between the two systems was identified and eliminated, ensuring perfect temporal alignment of the physiological events [13].

Furthermore, the raw continuous-wave radar data provided in this clinical dataset is not a direct 1D displacement signal. Instead, it is exported as complex In-Phase (I) and Quadrature (Q) components (radar_I and radar_Q), recorded in millivolts. Extracting the actual mechanical chest displacement from these raw components requires rigorous signal processing, including ellipse fitting to compensate for hardware imbalances (offset, gain, and phase errors), followed by arctangent demodulation [13].

By utilizing this vast, multifaceted clinical dataset as the primary source material, the subsequent methodology moves away from reliance on pre-packaged, mathematically simplified subsets. Instead, a custom data initialization and preprocessing pipeline was developed to independently extract, align, and manipulate these raw cardiopulmonary recordings, as detailed in the following section.

4.3 Custom Data Initialization

To fully leverage the physiological richness of the raw clinical dataset and address the architectural limitations identified in the reference framework, a completely custom data preprocessing pipeline was developed in MATLAB. Instead of relying on pre-segmented subsets, this script systematically extracts, cleans, demodulates, and windows the raw continuous data. The methodology is broken down into sequential processing blocks.

Initialization and Cohort Splitting

Out of the 30 available participants in the clinical dataset, a specific cohort of ten individuals (Subject IDs 11 through 20) was selected. To ensure robust model evaluation and strictly prevent any data leakage, the cohort was deterministically partitioned: Subjects 11 through 18 were allocated exclusively for training and validation, while Subjects 19 and 20 were isolated for testing. Furthermore, the segment length was deliberately initialized at 2048 points (as opposed to the standard 1024) to provide an extended temporal context.

```
inputBaseDir = 'C:\Users\Kourkouk\Documents\MyRawData';
outputBaseDir = 'C:\Users\Kourkouk\Documents\ProcessedDataset';
segmentLength = 2048;
downsampleFactor = 10;

% Output directory initialization
foldersToMake = {fullfile(outputBaseDir, 'trainVal', 'radar'), ...
                 fullfile(outputBaseDir, 'trainVal', 'ecg'), ...
                 fullfile(outputBaseDir, 'test', 'radar'), ...
                 fullfile(outputBaseDir, 'test', 'ecg')};
```

```

for i = 1:length(foldersToMake)
    if ~exist(foldersToMake{i}, 'dir'), mkdir(foldersToMake{i}); end
end

% Subject iteration and deterministic splitting
for subject = 11:20
    if subject <= 18
        destFolder = "trainVal";
    else
        destFolder = "test";
    end
    subjName = sprintf('GDN%04d', subject);
    subjPath = fullfile(inputBaseDir, subjName);
    if ~exist(subjPath, 'dir')
        warning(['File not found: ', subjName]);
        continue;
    end

```

Scenario Selection

Importantly, data from all five physiological scenarios—Resting, Valsalva, Apnea, Tilt Up, and Tilt Down—were included. Training the neural network on highly non-stationary states (such as the sudden shifts in blood pressure caused by orthostatic stress in the Tilt scenarios) ensures that the final architecture can generalize to real-world, dynamic cardiopulmonary events. The script actively filters the loaded files to include only these specific scenarios.

```

files = dir(fullfile(subjPath, '*.mat'));
scenarios = ["Resting", "Valsalva", "Apnea", "TiltUp", "TiltDown"];

for f = 1:length(files)
    fileName = files(f).name;
    containsScenario = false;
    currentScenario = "";

    for s = scenarios
        if contains(fileName, s, 'IgnoreCase', true)
            containsScenario = true;
            currentScenario = s;
            break;
        end
    end
    if ~{containsScenario}, continue; end

    data = load(fullfile(subjPath, fileName));

```

Signal Imputation and Phase Demodulation

Raw physiological recordings often contain non-finite artifacts (NaN, Inf) due to sensor disconnections. These were corrected using linear interpolation (`fillmissing`) to maintain temporal continuity. Subsequently, the physical displacement of the chest wall was extracted from the radar's raw In-Phase (I) and Quadrature (Q) channels. They were combined into a complex vector, and the physiological phase was extracted using an arctangent function, wrapped by an `unwrap` operation to prevent non-linear phase jumps.

```
% ECG Cleaning
ecg_raw = data.tfm_ecg1;
ecg_raw(~isfinite(ecg_raw)) = NaN;
ecg_raw = fillmissing(ecg_raw, 'linear');
ecg_raw(isnan(ecg_raw)) = 0;

% Radar I/Q Cleaning and Demodulation
radar_i_raw = fillmissing(data.radar_i, 'linear');
radar_q_raw = fillmissing(data.radar_q, 'linear');

radar_complex = radar_i_raw + 1i * radar_q_raw;
radar_raw = unwrap(angle(radar_complex));
```

Downsampling and Ground-Truth Filtering

The original signals, sampled at an exceptionally high frequency of 2000 Hz, contain redundant data points that exponentially increase computational overhead. Consequently, both signals were decimated by a factor of 10 (resulting in 200 Hz). A critical intervention followed: a digital bandpass filter (0.5 Hz to 40 Hz) was applied to the downsampled ECG. This effectively eliminates low-frequency baseline wander and high-frequency muscle/powerline noise, providing the neural network with a morphologically pristine ground-truth target.

```
% Decimation to 200 Hz
ecg_down = decimate(ecg_raw, downsampleFactor);
radar_down = decimate(radar_raw, downsampleFactor);

% Ground-truth bandpass filtering (0.5 Hz - 40 Hz)
ecg_down = bandpass(ecg_down, [0.5, 40], 200);
```

Extended Temporal Segmentation (2048 Points)

The final modification was the segmentation process. At a 200 Hz sampling rate, the chosen 2048-point array equates to 10.24 seconds of continuous physiological recording per segment. This extended sequence allows the Bidirectional LSTM layers to observe significantly more cardiac cycles within a single batch, improving its ability to adapt to transient heart rate changes and establish an accurate Inter-Beat Interval (IBI). The segments were then reshaped into 1×2048 arrays and saved for deep learning ingestion.

```
numSegments = floor(length(ecg_down) / segmentLength);

for k = 1:numSegments
    idx_start = (k-1)*segmentLength + 1;
    idx_end = k*segmentLength;

    ecg_segment = ecg_down(idx_start:idx_end);
    radar_segment = radar_down(idx_start:idx_end);

    baseName = sprintf('%s_%s_S%02d_%04d.mat', currentScenario, subject,
k);

    radarSavePath = fullfile(outputBaseDir, destFolder, 'radar', sprintf(
baseName, 'radar'));

    ecgSavePath = fullfile(outputBaseDir, destFolder, 'ecg', sprintf(
baseName, 'ecg'));
    dataEcg = reshape(ecg_segment, 1, segmentLength);
    save(ecgSavePath, 'dataEcg');

    dataRadar = reshape(radar_segment, 1, segmentLength);
    save(radarSavePath, 'dataRadar');
end
end
```

With the custom preprocessing pipeline successfully implemented, the resulting dataset provides a highly refined, temporally extended, and artifact-free foundation for deep learning ingestion. By transitioning away from the generic, pre-packaged subsets and utilizing this rigorously conditioned cohort, the subsequent experimental phase can focus purely on architectural optimization and algorithmic innovation. In the following chapter, this newly generated dataset serves as the primary experimental benchmark. It will be utilized to explore novel model configurations, evaluate the impact of the extended sequence length, and systematically refine the wavelet-augmented predictive model to ensure robust generalization across complex clinical scenarios.

Chapter 5

Custom Architecture Development and Implementation

5.1 Dynamic Data Loading and Scenario Filtering

Following the successful generation of the custom, extended-length cardiopulmonary dataset (as detailed in Chapter 4), the next step in the pipeline involves loading this data into the MATLAB workspace to train the core predictive model. To optimize memory utilization and facilitate structured data ingestion, the pipeline utilizes MATLAB's `signalDatastore` objects. These objects act as lightweight references pointing to the generated `.mat` files on the local drive, allowing for dynamic label extraction and subset filtering before actively consuming Random Access Memory (RAM).

Datastore Initialization and Label Extraction

The script initializes independent datastores for the radar inputs and the ECG targets across both the training/validation and testing directories. Concurrently, the categorical labels (representing the physiological scenarios) are extracted directly from the filenames using the `filenames2labels` function.

```
outputBaseDir = 'C:\Users\Kourkouk\Documents\ProcessedDataset';

radarTrainValDs = signalDatastore(fullfile(outputBaseDir,"trainVal","radar"));
;
ecgTrainValDs   = signalDatastore(fullfile(outputBaseDir,"trainVal","ecg"));

radarTestDs     = signalDatastore(fullfile(outputBaseDir,"test","radar"));
ecgTestDs       = signalDatastore(fullfile(outputBaseDir,"test","ecg"));

trainCats = filenames2labels(radarTrainValDs,'ExtractBefore','_radar');
testCats  = filenames2labels(radarTestDs,'ExtractBefore','_radar');
```

Scenario Filtering for Core Model Training

To establish a robust baseline for the core neural network, the initial training phase focuses on controlled physiological states (Resting, Apnea, Valsalva). Instead of permanently deleting the orthostatic stress data (Tilt Up, Tilt Down) from the hard drive, the script utilizes logical indexing and string matching (`contains`) to dynamically exclude any file containing the word "Tilt" from the active datastores and their corresponding label arrays.

This programmatic flexibility is crucial; it establishes a stable foundation for the core model's initial development, while retaining the entire dataset architecture for subsequent, advanced experimental phases (detailed in Chapter 6), where the extreme stress scenarios can be easily reintroduced by simply bypassing this filtering block.

```
tiltIdxTrain = contains(string(trainCats), 'Tilt', 'IgnoreCase', true);
radarTrainValDs = subset(radarTrainValDs, ~tiltIdxTrain);
ecgTrainValDs = subset(ecgTrainValDs, ~tiltIdxTrain);
trainCats(tiltIdxTrain) = [];

tiltIdxTest = contains(string(testCats), 'Tilt', 'IgnoreCase', true);
radarTestDs = subset(radarTestDs, ~tiltIdxTest);
ecgTestDs = subset(ecgTestDs, ~tiltIdxTest);
testCats(tiltIdxTest) = [];
```

Normalization, Splitting, and Memory Loading

With the desired scenarios isolated, the ECG target datastores are passed through a custom normalization transform to ensure consistent amplitude scaling. The input (radar) and target (ECG) datastores are then strictly synchronized using the `combine` function.

A critical divergence from the reference MathWorks implementation occurs during the data partitioning phase. In contrast to the deterministic data splitting often employed in baseline frameworks, initial exploratory training runs within this custom pipeline allowed for unconstrained random states to observe the model's behavior across varied subset distributions. However, to ensure strict scientific reproducibility and enable fair, direct comparisons between different architectural iterations (such as evaluating the impact of the Attention mechanism), setting a fixed random seed (e.g., `rng('default')`) or a specific integer) prior to the `splitlabels` function is formally recommended and adopted for the final evaluation phases.

The combined `trainValDs` is thus randomly partitioned into an 80% training subset and a 20% validation subset, utilizing the extracted `trainCats` labels to maintain proportional class stratification. Finally, because the custom dataset cohort has been downsampled and carefully segmented, it fits within standard memory limits. The `readall` command is explicitly invoked to

load the entire structured dataset into active RAM, drastically accelerating the data ingestion rate during the deep learning training loop.

```
ecgTrainValDs = transform(ecgTrainValDs, @helperNormalize);
ecgTestDs     = transform(ecgTestDs, @helperNormalize);

trainValDs = combine(radarTrainValDs, ecgTrainValDs);
testDs     = combine(radarTestDs, ecgTestDs);

splitIndices = splitlabels(trainCats, 0.8);
trainDs      = subset(trainValDs, splitIndices{1});
valDs        = subset(trainValDs, splitIndices{2});

trainData = readall(trainDs);
valData   = readall(valDs);
testData  = readall(testDs);
```

5.2 Core Model Architecture and Initial Training

While the previous chapter explored the pre-configured MathWorks reference framework, this section introduces the primary contribution of this thesis: a fully customized deep learning architecture designed from scratch. Building upon the theoretical insights of wavelet analysis, this core model introduces significant structural modifications to accommodate the extended 2048-point temporal sequences and to enhance regularization.

Architecture Definition and Full-Spectrum Wavelet Extraction

The defining characteristic of this initial custom architecture is its approach to feature extraction. While previous iterations explicitly discarded specific multiresolution levels to blindly filter out respiration, this core model utilizes a `modwtLayer` configured to extract and pass *all* detail levels (Levels 1 through 5) to the subsequent layers. By setting `SelectedLevels` to `1 : 5`, the network is forced to process the entire frequency spectrum of the mechanical signal. This establishes a baseline to observe how well the standard convolutional layers can intrinsically learn to separate the cardiac signature from respiratory and movement artifacts without forceful prior elimination.

Furthermore, to combat overfitting on the larger temporal windows, dropout layers (with a 30% dropout rate) were strategically integrated within the encoder blocks.

```
layers_attention = [
    sequenceInputLayer(1, MinLength=2048, Name="input")
```

```

modwtLayer('Level', 5, 'IncludeLowpass', false, 'SelectedLevels', 1:5, "
Wavelet", "sym4", Name="modwt")

flattenLayer(Name="flatten")
layerNormalizationLayer(Name="ln")
dropoutLayer(0.3, Name="drop1")

convolution1dLayer(64, 8, Padding="same", Stride=8, Name="conv1")
reluLayer(Name="relu1")
batchNormalizationLayer(Name="bn1")
dropoutLayer(0.3, Name="drop2")
maxPooling1dLayer(2, Padding="same", Name="pool1")

convolution1dLayer(32, 8, Padding="same", Stride=4, Name="conv2")
reluLayer(Name="relu2")
batchNormalizationLayer(Name="bn2")
maxPooling1dLayer(2, Padding="same", Name="pool2")

transposedConv1dLayer(32, 8, Cropping="same", Stride=4, Name="deconv1")
reluLayer(Name="relu3")
transposedConv1dLayer(64, 8, Cropping="same", Stride=8, Name="deconv2")

bilstmLayer(8, OutputMode="sequence", Name="bilstm")
fullyConnectedLayer(4, Name="fc1")
fullyConnectedLayer(2, Name="fc2")
fullyConnectedLayer(1, Name="fc_out")
];

```

Training Configuration and Loss Function

The training pipeline was strictly configured to handle the CTB (Channel-Time-Batch) data formats required by the modern trainnet API. The optimization process relies on the Adam solver with an initial learning rate of 0.001, executing over a maximum of 200 epochs. To manage the heavier computational load of the 2048-point sequences, a mini-batch size of 450 was selected, dynamically distributed across multiple CPU cores (ExecutionEnvironment="parallel").

The objective function driving the optimization is a custom implementation, waveletECGLoss, mathematically adapted to evaluate the Mean Squared Error across the extended sequence boundaries.

```

options = trainingOptions("adam",...
    MaxEpochs=200, MiniBatchSize=450, InitialLearnRate=0.001,...
    ValidationData={valData(:,1),valData(:,2)},...
    ValidationFrequency=40, Verbose=1, Shuffle="every-epoch",...
    OutputNetwork="best-validation", Plots="training-progress",...

```

```
InputDataFormats="CTB", TargetDataFormats="CTB", ...
ExecutionEnvironment="parallel");

lossFcn = @(y,t) waveletECGLoss(y,t);
```

Initial Training Dynamics and Convergence

The execution of the training loop revealed critical insights into the network’s learning behavior. As depicted in the training progress graph (Figure 5.1), the model rapidly minimizes the training loss during the initial epochs.

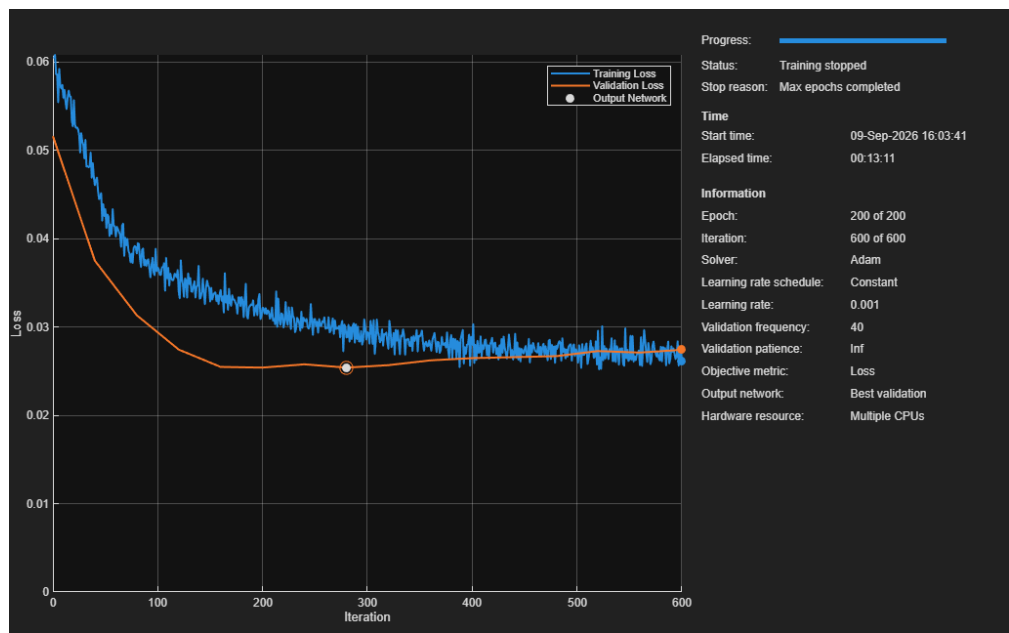


FIGURE 5.1: Training and Validation Loss history for the core model. The network architecture effectively utilizes the best-validation checkpointing to prevent overfitting as the validation curve plateaus.

However, a subtle divergence between the training and validation loss curves emerges mid-way through the process (around iteration 280). While the training loss continues to steadily decline, the validation loss plateaus and begins to show minor fluctuations. This behavior highlights the absolute necessity of the {OutputNetwork="best-validation" parameter, which successfully captured and retained the optimal network weights prior to the onset of overfitting, ensuring maximum generalization on unseen data.

Reconstruction Evaluation Across Physiological Scenarios

To qualitatively assess the generalization capabilities of this core architecture, the trained model was deployed to reconstruct signals across all five physiological scenarios from the unseen testing subset.

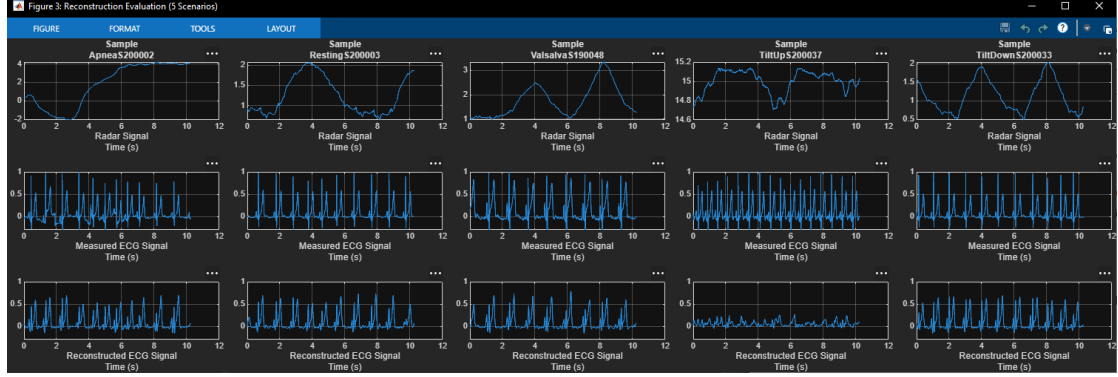


FIGURE 5.2: Signal reconstruction evaluation across five distinct physiological scenarios. The model demonstrates high fidelity in most states but struggles significantly with morphological accuracy during the *TiltUp* scenario.

As illustrated in Figure 5.2, the model exhibits remarkable accuracy in extracting the cardiac signature during the controlled and stationary scenarios, specifically Apnea, Resting, and Valsalva. In these states, the reconstructed ECG demonstrates high morphological fidelity and highly correlates with the ground-truth R-peaks, successfully tracking the complex temporal rhythm regardless of the underlying respiration.

Crucially, however, the model encounters a severe performance degradation during orthostatic stress scenarios, notably TiltUp and to a significant extent TiltDown (Figure 5.2, fourth column). In this highly dynamic state of orthostatic stress, the reconstructed signal’s amplitude drops drastically, and the morphological structure of the QRS complexes becomes chaotic and misaligned.

This specific failure highlights a fundamental limitation of passing the entire, unfiltered multiresolution spectrum (Levels 1:5) directly into the convolutional layers. When faced with extreme physiological noise, the standard layers lack the capability to dynamically prioritize the correct frequency bands. This critical observation serves as the primary motivation for Chapter 6, where targeted level exclusion and advanced attention mechanisms will be introduced to solve this exact problem.

Attention Layer Implementation: Properties and Initialization

The custom layer, WaveletChannelAttentionLayer, operates by evaluating the output channels generated by the MODWT layer. To prevent excessive computational overhead while learning these channel interactions, the layer initializes a small, two-layer perceptron bottleneck defined by a reduction ratio r (implemented as reductionRatio).

```
classdef WaveletChannelAttentionLayer < nnet.layer.Layer & nnet.layer.  
    Formattable  
    properties (Learnable)  
        Weights1  
        Bias1  
        Weights2  
        Bias2  
    end  
  
    properties  
        NumChannels  
        ReductionRatio  
    end  
  
    methods  
        function layer = WaveletChannelAttentionLayer(numChannels,  
reductionRatio, name)  
            layer.Name = name;  
            layer.Description = "1D SE Channel Attention";  
            layer.NumChannels = numChannels;  
  
            if nargin < 2 || isempty(reductionRatio)  
                reductionRatio = 2;  
            end  
            layer.ReductionRatio = reductionRatio;  
  
            hiddenDim = max(1, round(numChannels / reductionRatio));  
  
            layer.Weights1 = randn([hiddenDim, numChannels], 'single') * sqrt  
(2 / numChannels);  
            layer.Bias1 = zeros([hiddenDim, 1], 'single');  
  
            layer.Weights2 = randn([numChannels, hiddenDim], 'single') * sqrt  
(2 / hiddenDim);  
            layer.Bias2 = zeros([numChannels, 1], 'single');  
        end  
    end  
end
```

5.2.1 From Hardcoded Filtering to Dynamic Attention

The performance degradation observed in the *TiltUp* scenario directly exposes a fundamental flaw in traditional, empirically driven feature selection. In the reference MathWorks framework, the engineers employed a rigid heuristic: they manually configured the network to outright discard specific wavelet decomposition levels (such as Levels 1, 2, and the scaling coefficients) based on the assumption that respiratory and motion artifacts always reside within those exact frequency bands.

While this static heuristic filtering works adequately in highly controlled, stationary states (like Resting or Apnea), it is highly susceptible to failure in dynamic, real-world clinical environments. During extreme orthostatic stress (such as the Tilt scenarios), the physiological frequencies shift, and sudden macroscopic body movements introduce wideband noise. In such non-stationary conditions, statically dropping predefined levels might inadvertently discard crucial transient cardiac information or fail to filter out shifted motion artifacts.

To solve this, the network must abandon rigid heuristics and instead *learn* to dynamically evaluate the importance of each frequency band. Inspired by the seminal work of Hu et al. [4] on Squeeze-and-Excitation (SE) networks, a custom 1D Channel Attention mechanism was developed and integrated immediately after the `modwtLayer`. Unlike spatial attention which focuses on "where" the information is, this SE mechanism explicitly models channel interdependencies to dictate "what" frequency bands to attend to.

Attention Layer Implementation: Properties and Initialization

The custom layer, `WaveletChannelAttentionLayer`, operates by evaluating the output channels generated by the MODWT layer. To prevent excessive computational overhead while learning these channel interactions, the layer initializes a small, two-layer perceptron bottleneck defined by a reduction ratio r (implemented as `reductionRatio`).

```
classdef WaveletChannelAttentionLayer < nnet.layer.Layer & nnet.layer.  
    Formattable  
    properties (Learnable)  
        Weights1  
        Bias1  
        Weights2  
        Bias2  
    end  
  
    properties  
        NumChannels  
        ReductionRatio  
    end
```

```

methods
    function layer = WaveletChannelAttentionLayer(numChannels,
reductionRatio, name)
        layer.Name = name;
        layer.Description = "1D SE Channel Attention";
        layer.NumChannels = numChannels;

        if nargin < 2 || isempty(reductionRatio)
            reductionRatio = 2;
        end
        layer.ReductionRatio = reductionRatio;

        hiddenDim = max(1, round(numChannels / reductionRatio));

        layer.Weights1 = randn([hiddenDim, numChannels], 'single') * sqrt
(2 / numChannels);
        layer.Bias1 = zeros([hiddenDim, 1], 'single');

        layer.Weights2 = randn([numChannels, hiddenDim], 'single') * sqrt
(2 / hiddenDim);
        layer.Bias2 = zeros([numChannels, 1], 'single');
    end

```

Mathematical Formulation: Squeeze and Excitation

To mathematically define the layer's operation, let the multiresolution feature map extracted by the preceding MODWT layer be denoted as $\mathbf{X} \in \mathbb{R}^{C \times T}$, where $C = 5$ represents the number of wavelet channels (Levels 1 through 5) and $T = 2048$ is the sequence length.

During the forward pass (predict), the layer first performs the "**Squeeze**" operation. Since convolutional filters operate with local receptive fields, they lack global context. To resolve this, Global Average Pooling (GAP) is applied across the temporal dimension T to compress the sequence of each wavelet level into a single scalar value. This generates a channel descriptor $\mathbf{z} \in \mathbb{R}^C$ (represented by the variable *s_flat* in the implementation), whose c -th element is calculated as:

$$z_c = \frac{1}{T} \sum_{t=1}^T \mathbf{X}_c(t) \quad (5.1)$$

This statistic acts as a global summary of the energy present in that specific frequency band. Subsequently, the "**Excitation**" operation captures the channel-wise dependencies. This is

achieved by passing the descriptor $\mathbf{z}$ through the learnable bottleneck fully connected network to yield the intermediate hidden state $\mathbf{h}$ and the final attention vector $\mathbf{s}$ (represented by a_flat):

$$\mathbf{h} = \delta(\mathbf{W}_1 \mathbf{z} + \mathbf{b}_1) \quad (5.2)$$

$$\mathbf{s} = \sigma(\mathbf{W}_2 \mathbf{h} + \mathbf{b}_2) \quad (5.3)$$

where δ denotes the ReLU activation function, σ denotes the Sigmoid activation function, $\mathbf{W}_1 \in \mathbb{R}^{\frac{C}{r} \times C}$ is the weight matrix of the dimensionality-reduction layer (Weights1), and $\mathbf{W}_2 \in \mathbb{R}^{C \times \frac{C}{r}}$ is the weight matrix of the dimensionality-increasing layer (Weights2). The Sigmoid function ensures that the final output $\mathbf{s}$ is a vector of continuous attention weights strictly bounded between 0 and 1, representing the dynamic relevance of each wavelet channel.

```
function Z = predict(layer, X)
    if isa(X, 'dlarray')
        fmt = dims(X);
        X_data = stripdims(X);
    else
        fmt = '';
        X_data = X;
    end

    C = size(X_data, 1);

    % Squeeze Operation (Eq. 1)
    s = mean(X_data, 2);
    s_flat = reshape(s, C, []);

    % Excitation Operations (Eq. 2 & Eq. 3)
    h = relu(layer.Weights1 * s_flat + layer.Bias1);
    a_flat = sigmoid(layer.Weights2 * h + layer.Bias2);
```

Dynamic Re-weighting

In the final step, the network utilizes the learned activations to dynamically calibrate the input features. The original multiresolution feature map is re-weighted by performing an element-wise multiplication (broadcasting) with the attention mask $\mathbf{s}$:

$$\tilde{\mathbf{X}}_c = \mathbf{s}_c \cdot \mathbf{X}_c \quad (5.4)$$

where $\tilde{\mathbf{X}} = [\tilde{\mathbf{X}}_1, \tilde{\mathbf{X}}_2, \dots, \tilde{\mathbf{X}}_C]$ becomes the optimized output tensor (Z_data). If the network determines that a specific wavelet level contains overwhelming motion artifacts during a particular 10.24-second window, its corresponding weight s_c is driven towards zero, effectively muting that channel. Conversely, channels rich in cardiac information are amplified.

```

    % Dynamic Re-weighting (Eq. 4)
    szA = size(X_data);
    szA(2) = 1;

    a = reshape(a_flat, szA);
    Z_data = X_data .* a;

    if ~isempty(fmt)
        Z = dldarray(Z_data, fmt);
    else
        Z = Z_data;
    end
end
end
end

```

By integrating this mathematical calibration into the WaveletChannelAttentionLayer, the model successfully evolves from a static, empirically filtered system into an adaptive, context-aware network capable of dynamically suppressing non-stationary noise.

5.2.2 Spectral Regularization via Wavelet-Domain Loss

While the integration of the Channel Attention mechanism (detailed in Section 5.2.1) significantly optimizes the input feature maps, the objective function guiding the network's optimization must also be refined. Standard deep learning regression tasks typically rely on the time-domain Mean Squared Error (MSE) to compute the gradient updates. However, relying solely on time-domain MSE presents a well-documented challenge in biomedical signal processing: the network often learns to simply output a highly smoothed, averaged waveform to minimize point-to-point temporal distances, effectively "drawing" an approximation of the signal while failing to reconstruct critical, high-frequency transients like the sharp R-peaks of the QRS complex.

To force the network to learn actual structural and physiological information rather than merely mimicking a temporal shape, a custom spectral regularization technique was introduced. This was achieved by developing a custom objective function, `waveletECGLoss`, which computes the error not only in the time domain but also simultaneously in the wavelet domain.

Mathematical Formulation of the Dual-Domain Loss

During the forward pass, the custom loss function intercepts the predicted ECG signal ($\hat{y}$) and the ground-truth ECG target (y). Before computing the final error, both signals are mathematically transformed into the multiresolution domain using the Maximal Overlap Discrete Wavelet Transform (MODWT).

By extracting the wavelet coefficients for both the prediction and the target, the loss function can explicitly penalize discrepancies across specific frequency bands. The total loss $\mathcal{L}_{total}$ is thus formulated as a weighted combination of the temporal loss and the mean spectral loss:

$$\mathcal{L}_{total} = \text{MSE}(y, \hat{y}) + \lambda \frac{1}{L} \sum_{j=1}^L \text{MSE}(\mathbf{W}_j(y), \mathbf{W}_j(\hat{y}))$$

where:

- y and $\hat{y}$ represent the true and predicted 2048-point ECG sequences, respectively.
- $\mathbf{W}_j(\cdot)$ denotes the MODWT coefficients at decomposition level j .
- L is the total number of wavelet decomposition levels analyzed, with the spectral penalty computed as the mean MSE across all levels.
- λ is a tunable regularization hyperparameter that controls the influence of the spectral penalty.

Physiological Impact on Network Convergence

By explicitly calculating the loss across these detail levels, the neural network is severely penalized if it ignores the high-frequency components that define the cardiac micro-motions. It forces the Bidirectional LSTM and the transposed convolutional decoder to actively reconstruct the true physiological frequencies, aligning the spectral energy of the predicted signal with that of the clinical ground truth. This dual-domain optimization ensures that the resulting waveforms maintain their diagnostic morphology, preventing the over-smoothing artifacts traditionally associated with pure temporal regression models.

Implementation and Hyperparameter Optimization

The custom loss function is implemented to intercept the predicted and target sequences, strictly enforcing real-number operations to prevent complex-plane artifacts during the gradient calculations. The `dlmodwt` function is utilized to dynamically compute the wavelet coefficients during the network’s forward pass, explicitly extracting the first five detail levels for comparison.

```
function loss = waveletECGLoss(y, t, lambda)
    y = real(y);
    t = real(t);

    lossECG = mean((y-t).^2, 'all');

    waveletY = real(dlmodwt(y, [], [], 5));
    waveletT = real(dlmodwt(t, [], [], 5));

    waveletY = waveletY(1:5, :, :, :);
    waveletT = waveletT(1:5, :, :, :);

    lossWavelet = mean((waveletY - waveletT).^2, 'all');

    loss = lossECG + lambda * lossWavelet;
end
```

A mathematically critical component of this objective function is the regularization hyperparameter, λ . This scalar dictates the precise trade-off between temporal smoothness (Time-domain MSE) and morphological structural accuracy (Wavelet-domain MSE). In deep learning architectures, dynamically balancing a dual-objective loss function is a notoriously complex optimization challenge. An improperly tuned λ can either negate the spectral penalty entirely, or conversely, overwhelm the gradient updates, preventing the network from achieving baseline temporal convergence.

Rather than relying on arbitrary heuristics or unverified assumptions—a fundamental limitation observed in the reference framework’s empirically predetermined level exclusion—the determination of λ in this thesis was approached through a systematic hyperparameter sweep. To rigorously determine the optimal value and strictly prevent data leakage, this grid search was performed exclusively on the Validation subset. The model was trained incorporating both the Attention Layer and the Multi-loss mechanism across a discrete set of values: $\lambda \in \{0.1, 0.2, 0.3, 0.4, 0.5, 1.0\}$.

The performance of each configuration was subsequently evaluated using the Mean Squared Error (MSE) on the unseen validation data, alongside the required training time. The quantitative results of this experimental sweep are summarized in Table 5.1.

TABLE 5.1: Quantitative evaluation of the spectral regularization hyperparameter (λ) evaluated on the Validation Set to prevent data leakage.

Regularization Weight (λ)	Mean Validation MSE	Training Time (min)
0.1	0.025370	12:51
0.2	0.024586	13:43
0.3	0.024631	13:20
0.4	0.024260	11:52
0.5	0.024644	12:34
1.0	0.025355	13:06

As demonstrated in Table 5.1, setting $\lambda = 0.4$ yielded the lowest Mean Validation MSE (0.024260). This indicates the optimal mathematical balance between temporal smoothness and morphological structural accuracy for this dataset, without biasing the model to the final testing data.

Furthermore, this quantitative metric aligns directly with the rigorous qualitative evaluation of the reconstructed waveforms. Visual inspection confirmed that lower values ($\lambda \leq 0.2$) failed to adequately sharpen the transient QRS complexes, whereas setting the penalty too high ($\lambda = 1.0$) introduced excessive, erratic high-frequency noise into the signal. Consequently, $\lambda = 0.4$ was firmly locked as the permanent hyperparameter for the finalized architecture before any evaluation on the unseen Test Set was conducted.

5.3 Core Model Finalization and Performance Evaluation

With the integration of both the dynamic Squeeze-and-Excitation Channel Attention (Section 5.2.1) and the dual-objective spectral loss function (Section 5.2.2), the custom deep learning architecture is fully realized. This upgraded network represents a significant evolution from the initial implementation, transitioning from a rigidly filtered time-domain regressor to a context-aware, spectrally regulated model. This finalized architecture will serve as the robust *baseline model* against which future experimental iterations will be measured.

Training Execution and Quantitative Evaluation

The complete architecture (layers_attention) was compiled and trained using the optimal $\lambda = 0.4$ for the custom objective function. Following the training phase, the model was immediately deployed to evaluate the unseen testing dataset using the minibatchpredict function. To quantitatively assess the performance, the Mean Squared Error (MSE) was computed across all reconstructed sequences.

```
disp('=== Training: Wavelet Attention ===');
```

```

[net_attention, ~] = trainnet(trainData(:,1), trainData(:,2),
    layers_attention, lossFcn, options);

disp('=== Testing: Model Evaluation ===');
ecgTestRe_att = minibatchpredict(net_attention, testData(:,1),
    InputDataFormats="CTB", ExecutionEnvironment="cpu");
ecgTestReCell_att = squeeze(mat2cell(ecgTestRe_att, 1, 2048, ones(size(
    testData,1),1)));
lossAttModel = cellfun(@mse, ecgTestReCell_att, testData(:,2));

disp(['Mean Test MSE: ', num2str(mean(lossAttModel))]);

figure('Name', 'MSE Distribution');
histogram(lossAttModel, 'Normalization', 'probability', 'BinWidth', 0.002, '
    FaceColor', [0 0.4470 0.7410], 'FaceAlpha', 0.7);
ylabel("Probability"); xlabel("Test MSE");
title("Performance: Wavelet Attention Model"); grid on;

```

As illustrated in the training progress graph (Figure 5.3), the model demonstrates highly stable convergence. The best-validation checkpoint successfully halts the training near iteration 320, capturing the optimal network weights right before the validation loss begins to subtly diverge from the training loss, effectively preventing overfitting.

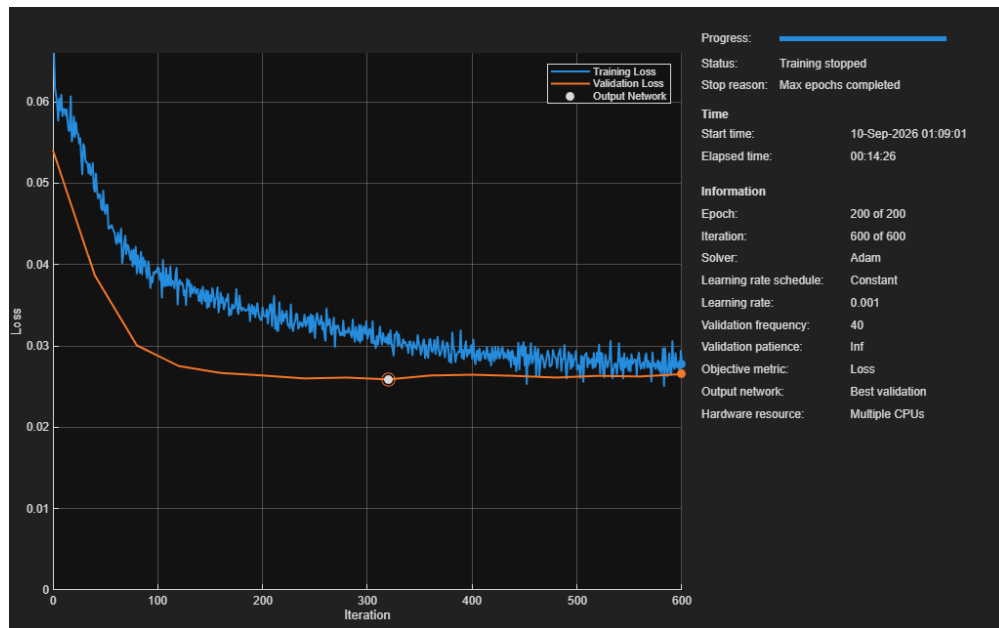


FIGURE 5.3: Training and Validation Loss history for the finalized baseline model (with Attention and Wavelet Loss). Convergence is smooth, with optimal weights extracted via validation patience.

Qualitative Visualization of Random Samples

To qualitatively validate the predictive capabilities of the baseline model, a visualization block was implemented to extract and plot random test samples across all five physiological categories.

```
disp('=== Generating 5 Random Sample Plots ===');
targetCategories = ["Apnea", "Resting", "Valsalva", "TiltUp", "TiltDown"];
selectedIndices = zeros(1, length(targetCategories));

for i = 1:length(targetCategories)
    currentCat = targetCategories(i);
    catIndices = find(testCats == currentCat);
    if ~isempty(catIndices)
        selectedIndices(i) = catIndices(randi(length(catIndices))));
    end
end

selectedIndices(selectedIndices == 0) = [];

if ~isempty(selectedIndices)
    figure('Name', 'Reconstruction Evaluation (5 Scenarios)', 'Color', [0.15
    0.15 0.15]);
    helperPlotData(testDs, selectedIndices, net_attention);
else
    disp('No samples found for visualization.');
```

To maintain visual consistency and enable a direct overlay of the temporal waveforms, the custom script leverages an adapted version of the helperPlotData function, originally utilized in the MathWorks framework.

```
function helperPlotData(DS,Indices,net1,net2)
arguments
    DS
    Indices
    net1 = []
    net2 = []
end
fs = 200;
N = numel(Indices);
M = 2;
if ~isempty(net1)
    M = M + 1;
end
if ~isempty(net2)
    M = M + 1;
```

```

end
tiledlayout(M, N, 'Padding', 'none', 'TileSpacing', 'compact');
for i = 1:N
    idx = Indices(i);
    ds = subset(DS,idx);
    [~,name,~] = fileparts(ds.UnderlyingDatastores{1}.Files{1});
    data = read(ds);
    radar = data{1};
    ecg = data{2};
    t = linspace(0,length(radar)/fs,length(radar));

    nexttile(i)
    plot(t,radar)
    title(["Sample",regexprep(name, {'_', 'radar'}, ' ')])
    xlabel(["Radar Signal", "Time (s)"])
    grid on

    nexttile(N+i)
    plot(t,ecg)
    xlabel(["Measured ECG Signal", "Time (s)"])
    ylim([-0.3,1])
    grid on

    if ~isempty(net1)
        nexttile(2*N+i)
        y = minibatchpredict(net1,radar,InputDataFormats="CTB");
        plot(t,y)
        grid on
        ylim([-0.3,1])
        xlabel(["Reconstructed ECG Signal", "Time (s)"])
    end
    if ~isempty(net2)
        nexttile(3*N+i)
        y = minibatchpredict(net2,radar,InputDataFormats="CTB");
        hold on
        plot(t,y)
        hold off
        grid on
        ylim([-0.3,1])
        xlabel(["Reconstructed ECG Signal", "with modwtLayer", "Time (s)"])
    end
end
end
set(gcf, 'Position', [0 0 300*N, 150*M])
end

```

Reconstruction Analysis and Scenario Transition

The resulting visualizations (Figure 5.4) highlight both the strengths and the remaining limitations of the current architecture.

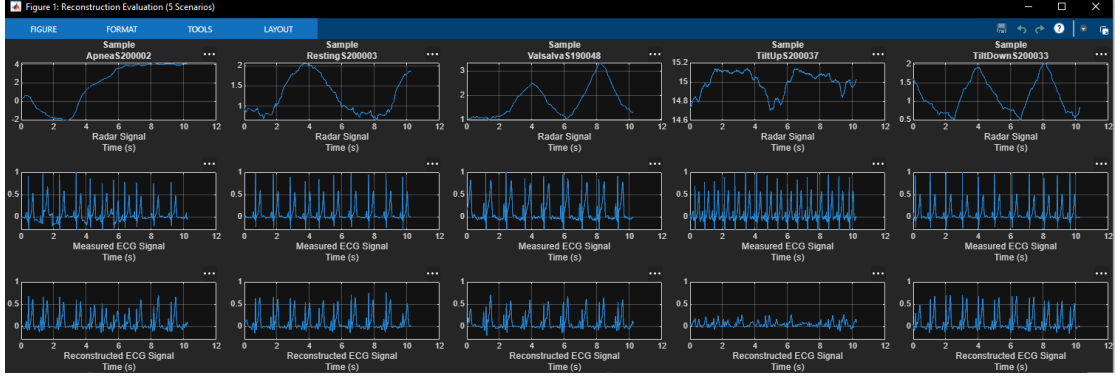


FIGURE 5.4: Signal reconstruction of the finalized baseline model across all five scenarios. The attention mechanism and spectral loss yield sharper R-peaks in stationary states, but orthostatic stress scenarios remain highly distorted.

In the first three scenarios (*Apnea*, *Resting*, *Valsalva*), the integration of the Wavelet Loss and Channel Attention demonstrates a tangible improvement. Compared to the initial model (Section 5.2), the network manages to capture the abrupt, sharp peaks of the true ECG with slightly higher precision, rather than generating overly smoothed curves.

However, the visual evidence confirms that the extreme orthostatic stress scenarios (*TiltUp* and *TiltDown*) continue to introduce a massive influx of complex, wideband noise and radical baseline shifts. While the attention mechanism attempts to re-weight the channels, the sheer volume of non-stationary artifacts physically overlapping with the cardiac frequencies causes the reconstructed signal to lose amplitude and morphological coherence.

Experimental Rationale for Chapter 6: Variable Isolation

While the finalized model demonstrates robust reconstruction capabilities in stationary and semi-stationary states, its susceptibility to severe orthostatic stress highlights a fundamental challenge in contactless sensing. The inclusion of the Tilt scenarios introduces such a high degree of variance and non-stationary macroscopic noise that it inherently skews the overall quantitative metrics (such as the Mean MSE). If these extreme stress scenarios remain in the dataset during fine-grained hyperparameter tuning, it becomes scientifically impossible to determine whether a specific architectural modification genuinely improved the network's intrinsic learning capacity or simply happened to filter a random motion artifact.

Therefore, to conduct rigorous, controlled ablation studies and structural experiments in Chapter 6, the dataset will be strategically restricted strictly to the three primary scenarios (*Apnea*,

Resting, Valsalva). By isolating the dataset to these foundational states, the upcoming experiments can accurately quantify the impact of architectural variations in a stable environment. Addressing the chaotic motion artifacts observed in the Tilt scenarios requires dedicated cancellation algorithms, which falls beyond the scope of this baseline architectural optimization and is reserved for future research.

Chapter 6

Further Research & Experimental Investigation

As concluded in Chapter 5, the inclusion of extreme orthostatic stress scenarios (TiltUp, Tilt-Down) introduces excessive non-stationary noise that inherently skews quantitative metrics. To conduct rigorous, controlled architectural experiments, the dataset for this chapter is strictly isolated to three stable scenarios: *Apnea*, *Resting*, and *Valsalva*.

To establish a definitive comparative baseline, the finalized core model from Section 5.3 was evaluated exclusively on this isolated dataset. This baseline evaluation yielded a **Mean Test MSE of 0.04374**.

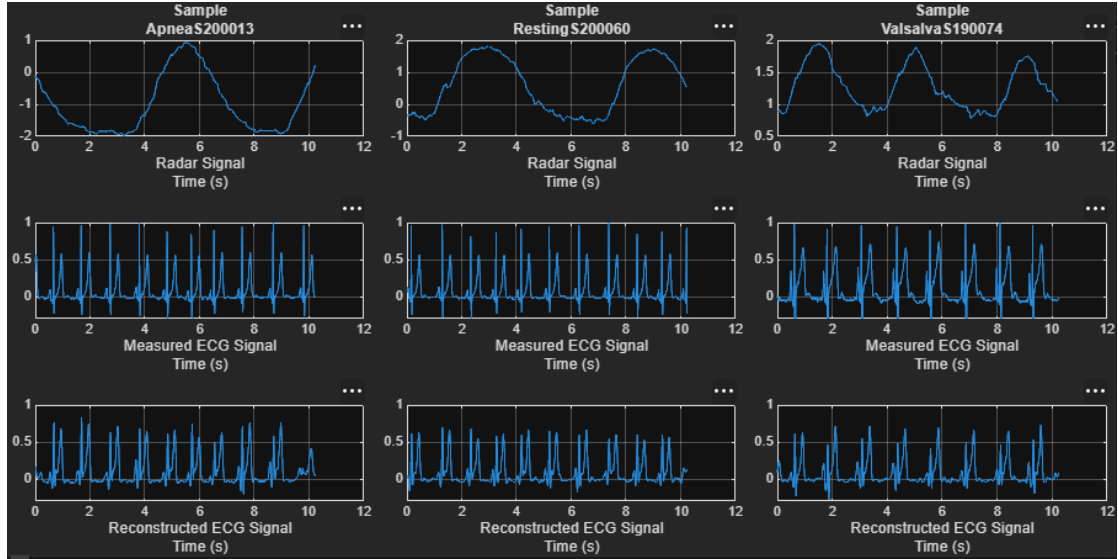


FIGURE 6.1: Baseline signal reconstruction across the isolated scenarios (Apnea, Resting, Valsalva). This visual benchmark highlights the current core limitation: the morphological flattening of the reconstructed QRS complexes.

Figure 6.1 serves as the fixed visual anchor for this chapter. While temporal rhythm tracking is highly stable, the network struggles to reconstruct the full amplitude of the R-peaks.

The subsequent sections in this chapter present independent, targeted experimental architectures. Each proposed mechanism is objectively evaluated against this exact quantitative (MSE: 0.04374) and qualitative baseline to determine its genuine impact on morphological fidelity.

6.1 Addressing Morphological Flattening: The Need for Temporal Awareness

As visually established in Figure 6.1, while the integration of the 1D Squeeze-and-Excitation mechanism successfully mitigates out-of-band noise, a critical structural limitation persists: the systematic flattening of the QRS complexes. Even within the controlled, semi-stationary scenarios (Apnea, Resting, Valsalva), the network struggles to reach the true amplitude of the R-peaks.

The learned mask is expected to assign higher weights of the current feature extraction. The existing Channel Attention mechanism effectively identifies *which* multiresolution frequency bands possess the highest signal-to-noise ratio, but it applies this learned weight globally across the entire 10.24-second sequence. Because the attention block lacks intrinsic temporal awareness, it cannot distinguish the millisecond-long occurrence of a cardiac contraction from the inter-beat interval. Consequently, the high-frequency transients are mathematically averaged and smoothed by the surrounding baseline.

To address this, the first experiment draws structural inspiration from Dual Attention Networks and 1-Dimensional Convolutional Block Attention Modules (1D-CBAM) utilized in standard, contact-based ECG processing. In traditional electrocardiology, cascading a Channel Attention block with a Spatial (Temporal) Attention block allows the network to learn both *what* to attend to and *when*. Adapting this architecture to the contactless radar domain, this section introduces a localized Temporal Attention mechanism immediately following the wavelet channel evaluation.

The hypothesis is strictly defined: by utilizing a 1D convolutional layer to scan the temporal axis and compute local variance across the wavelet channels, the network can learn to generate a dynamic temporal mask. This mask aims to selectively assign higher weights to the exact temporal indices of the R-peaks, theoretically maximizing their relative Signal-to-Noise Ratio (SNR) by preserving the peak integrity while actively suppressing the surrounding baseline noise, without violating the shift-invariance of the MODWT layer. The viability of this exploratory mechanism will be objectively determined by a singular criterion: its ability to visibly sharpen the QRS complexes and reduce the baseline Test MSE below the established 0.04374 threshold.

6.1.1 Mathematical Formulation of Temporal Attention

To computationally achieve temporal awareness without compressing the temporal dimension, the proposed mechanism leverages a 1D spatial attention paradigm. Let $\mathbf{X} \in \mathbb{R}^{C \times T}$ represent the multiresolution feature map outputted by the preceding Channel Attention block, where $C = 5$ denotes the number of wavelet levels and $T = 2048$ denotes the sequence length.

First, the mechanism aggregates the channel information at each temporal index t by applying average-pooling and max-pooling operations exclusively across the channel axis. This generates two distinct 1D temporal vectors that summarize the localized baseline energy and the peak intensity, respectively:

$$\mathbf{X}_{avg}(t) = \frac{1}{C} \sum_{c=1}^C \mathbf{X}_c(t), \quad \mathbf{X}_{max}(t) = \max_{c \in \{1, \dots, C\}} \mathbf{X}_c(t) \quad (6.1)$$

These independent temporal summaries are subsequently concatenated to form a composite feature descriptor $\mathbf{X}_{pool} \in \mathbb{R}^{2 \times T}$. To learn the temporal dependencies and identify the transient peaks, this descriptor is processed through a standard 1-Dimensional Convolution layer with a fixed kernel size $k = 7$. A sigmoid activation function (σ) is then applied to generate a continuous attention mask $\mathbf{M} \in \mathbb{R}^{1 \times T}$, bounded between 0 and 1:

$$\mathbf{M} = \sigma\left(\mathbf{W}^{1 \times 7} * [\mathbf{X}_{avg}; \mathbf{X}_{max}] + \mathbf{b}\right) \quad (6.2)$$

Finally, this temporal mask is broadcasted across the channel dimension and applied to the original input tensor via element-wise multiplication ($\otimes$). This operation actively suppresses the inter-beat temporal intervals (driving weights towards 0) while preserving the instantaneous QRS transients (driving weights towards 1):

$$\mathbf{Z} = \mathbf{X} \otimes \mathbf{M} \quad (6.3)$$

6.1.2 Software Implementation (Custom Layer)

To integrate this mathematical formulation into the MATLAB deep learning pipeline, a custom network layer, TemporalAttentionLayer, was developed. The layer natively processes formatted darray objects, leveraging dynamic dimension resolution to ensure structural compatibility with the preceding modwtLayer and the modern trainnet API.

```
classdef TemporalAttentionLayer < nnet.layer.Layer & nnet.layer.Formattable
    properties (Learnable)
        Weights
        Bias
    end
end
```

```

end

methods
    function layer = TemporalAttentionLayer(name)
        layer.Name = name;
        layer.Description = "1D Temporal Attention (CBAM)";

        % Explicit initialization with 'SCU' format
        % S = Spatial (Time), C = Channels (2), U = Unspecified (1 Filter
    )
        layer.Weights = dldarray(randn([7, 2, 1], 'single') * sqrt(2/14),
'SCU');
        layer.Bias = dldarray(zeros([1, 1, 1], 'single'), 'U');
    end

    function Z = predict(layer, X)
        % 1. Dynamically locate the Channel dimension
        cDim = finddim(X, 'C');
        if isempty(cDim), cDim = 1; end

        % 2. Channel-wise pooling (Equation 6.1)
        avg_pool = mean(X, cDim);
        max_pool = max(X, [], cDim);

        % 3. Concatenation along the Channel dimension
        concat_dl = cat(cDim, avg_pool, max_pool);

        % 4. 1D Convolution (Equation 6.2)
        conv_out = dlconv(concat_dl, layer.Weights, layer.Bias, 'Padding'
, 'same');

        % 5. Mask Generation (Equation 6.2)
        attention_mask = sigmoid(conv_out);

        % 6. Element-wise Masking (Equation 6.3)
        Z = X .* attention_mask;
    end
end
end

```

LISTING 6.1: Implementation of the 1D Temporal Attention Layer.

The algorithmic execution during the forward pass (the predict method) is structured into four critical computational stages to ensure numerical stability and strict dimensional consistency:

- **Dynamic Dimensionality Resolution:** Instead of stripping data formats, the layer dynamically identifies the structural layout of the input tensor using the finddim function.

By explicitly locating the channel dimension ('C'), the layer becomes robust against internal dimensional permutations applied by MATLAB's batching engine, ensuring that subsequent operations are applied to the correct axis.

- **Formatted Feature Aggregation:** The spatial pooling operations (mean and max) and the subsequent concatenation (cat) are executed directly on the formatted tensor along the identified channel axis. This approach intrinsically preserves the deep learning dimensional tags, resulting in a concatenated descriptor guaranteed to feature exactly two channels without requiring manual tensor reshaping or re-formatting.
- **Strictly Formatted Convolution and Initialization:** To prevent tensor mismatch during the 1D convolution, the learnable weights are explicitly initialized as a formatted darray using the 'SCU' label (Spatial, Channel, Unspecified). This guarantees that the dlconv function flawlessly aligns the weights with the input channels. The spatial kernel is set to size 7, and the Padding="same" parameter is enforced to maintain the precise 2048-point temporal alignment. Furthermore, He Initialization is utilized to maintain stable gradient flow.
- **Dynamic Mask Generation and Temporal Re-weighting:** The convolutional output is passed through a sigmoid activation, bounding the continuous temporal mask within the $[0, 1]$ interval. Element-wise multiplication broadcasts this mask across the original input. This dynamically re-weights the data: temporal indices containing physiological peaks approach a weight of 1, preserving their amplitude, while indices dominated by baseline noise are attenuated toward 0, dramatically improving the localized Signal-to-Noise Ratio (SNR).

6.1.3 Experimental Results & Evaluation

The augmented architecture, incorporating the Dual Attention mechanism, was trained and evaluated strictly under the established baseline conditions (Apnea, Resting, Valsalva). As depicted in the training progress (Figure 6.2), the integration of the custom Temporal Attention layer maintained gradient stability. The model converged smoothly without introducing erratic loss fluctuations or overfitting artifacts.

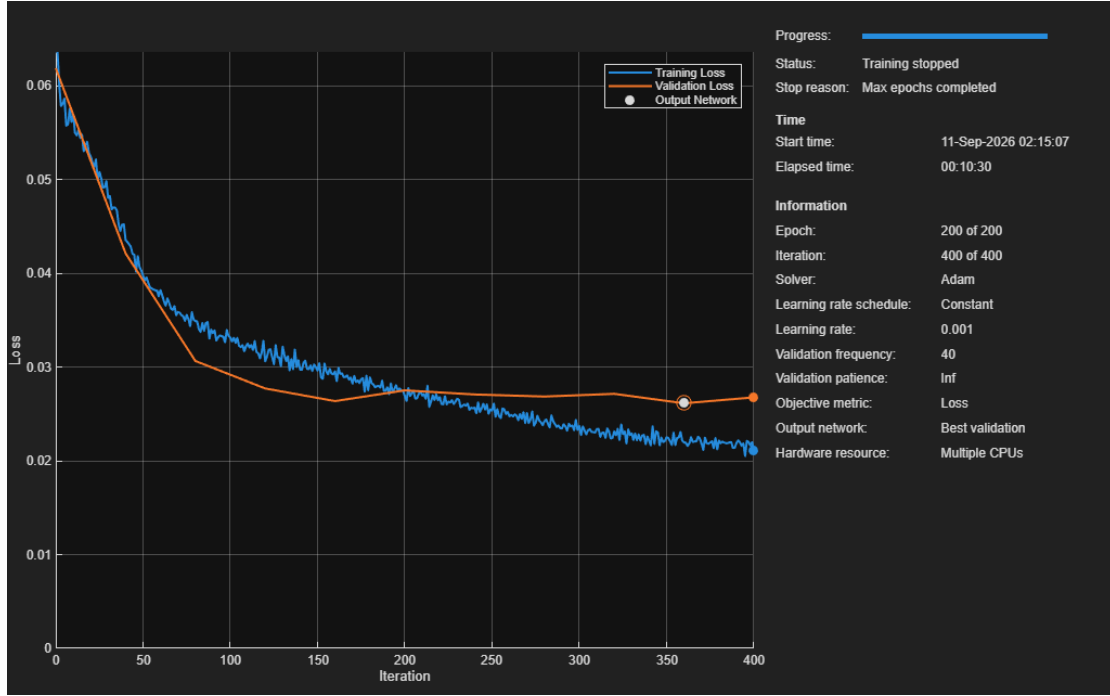
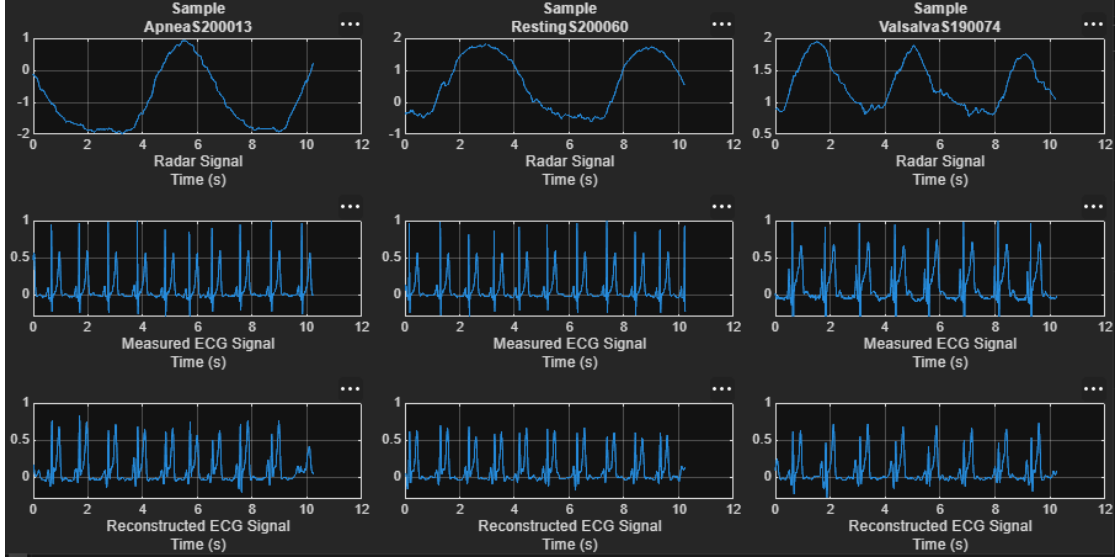


FIGURE 6.2: Training and validation loss convergence for the Dual Attention architecture. The learning process remains stable, indicating that the temporal mask does not disrupt gradient flow.

Quantitatively, the application of temporal re-weighting reduced the Mean Test MSE to 0.043133. Compared to the established baseline of 0.04374, this represents a marginal quantitative improvement of approximately 1.4%. However, the practical efficacy of the model must be evaluated through morphological analysis. To facilitate a direct objective assessment, Figure 6.3 juxtaposes the baseline reconstruction against the new experimental output.

Baseline Architecture (No Temporal Attention)



Experimental Architecture (With Temporal Attention)

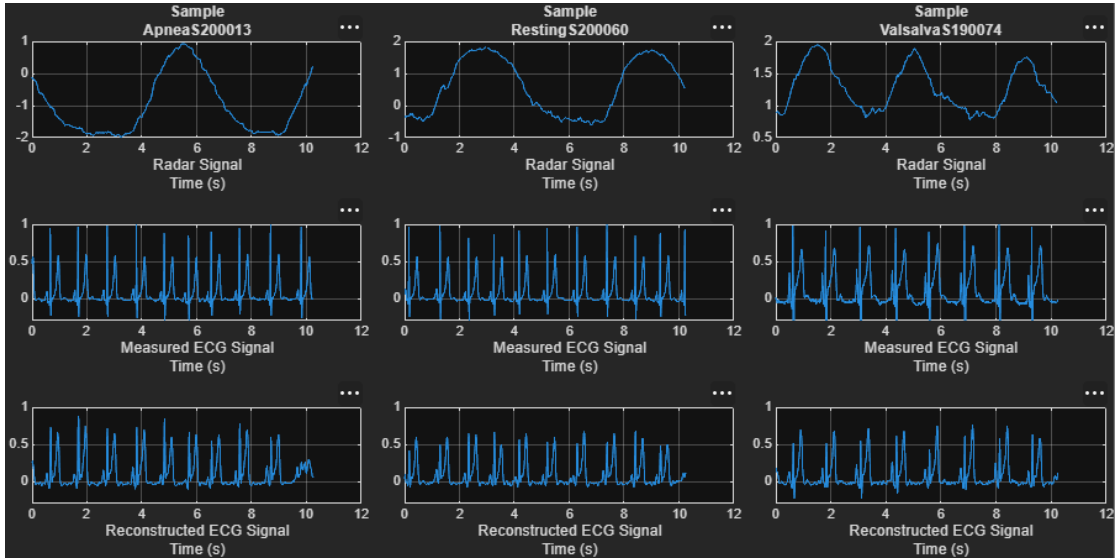


FIGURE 6.3: Direct visual comparison between the baseline model (top) and the Dual Attention model (bottom). The morphological flattening of the QRS complexes remains practically unchanged.

A critical visual inspection of Figure 6.3 refutes the initial hypothesis. The expectation was that the temporal mask would selectively maximize the local Signal-to-Noise Ratio (SNR) and significantly sharpen the R-peaks. In reality, the visual differences between the two outputs are virtually indistinguishable. The network still systematically fails to reconstruct the true amplitude of the QRS complexes, maintaining the smoothed, rounded approximations observed in the baseline model.

In conclusion, this experiment demonstrates that the straightforward cross-domain adaptation of 1D-CBAM principles is insufficient to resolve the morphological flattening inherent in contactless radar reconstruction. The persistence of this attenuation suggests that the limitation is not merely a lack of temporal awareness, but rather a deeper structural inability of standard convolutional decoders to mathematically synthesize sharp, high-amplitude electrical transients from non-linear macroscopic mechanical vibrations.

6.2 From Attenuation to Translation: Local Cross-Attention

The marginal improvements observed in the previous experiment dictate a fundamental shift in the structural approach. As concluded in Section 6.1.3, applying a temporal mask between 0 and 1 can effectively attenuate inter-beat noise, but it mathematically cannot synthesize high-frequency transients (such as the R-peaks) if those frequencies are inherently absent or severely dampened in the mechanical radar signal. A masking mechanism simply scales existing features; it does not translate them.

This observation motivates the hypothesis that reconstructing sharp electrical transients may require a more expressive cross-domain mapping mechanism rather than temporal re-weighting alone. Inspired by transformer architectures used in sequence-to-sequence modeling, this section proposes a Local Radar-to-ECG Wavelet Cross-Attention layer.

Unlike standard self-attention, this mechanism introduces a globally learnable set of query vectors that can adapt during training to identify radar features associated with the target ECG representation. The incoming multiresolution radar signal acts as the Keys and Values. By extracting local temporal windows, the network learns to correlate specific mechanical chest movements (Keys) with the ideal electrical patterns (Queries). When a match occurs, the mechanism learns which local radar features should contribute more strongly to each output channel and adds it to the signal via a residual connection, theoretically bridging the morphological gap between the two domains.

6.2.1 Mathematical Formulation of Local Cross-Attention

Let $\mathbf{X} \in \mathbb{R}^{C \times T}$ denote the input multiresolution radar tensor, where C is the number of wavelet channels and T is the temporal sequence length. To capture local mechanical dynamics, a sliding window of radius R is applied at each temporal index t , yielding a local temporal patch $\mathbf{X}^{(t)} \in \mathbb{R}^{C \times W}$, where the window size is $W = 2R + 1$.

The radar patches are mapped into Keys ($\mathbf{K}$) and Values ($\mathbf{V}$) using learnable weights ($\mathbf{W}_K$, $\mathbf{W}_V$) and biases. To preserve spatial identity within the window, a Channel Embedding ($\mathbf{E}_C$)

is added to the Keys:

$$\mathbf{K}^{(t)} = \mathbf{W}_K \mathbf{X}^{(t)} + \mathbf{E}_C, \quad \mathbf{V}^{(t)} = \mathbf{W}_V \mathbf{X}^{(t)} \quad (6.4)$$

Simultaneously, a globally learnable Query matrix $\mathbf{Q} \in \mathbb{R}^{E \times D_k}$ is initialized, where E represents the target ECG wavelet channels and D_k is the projection dimension. The attention scores are computed using the scaled dot-product between the ideal ECG Queries and the local radar Keys, followed by a softmax activation to ensure the weights sum to 1:

$$\mathbf{A}^{(t)} = \text{Softmax} \left(\frac{\mathbf{Q}(\mathbf{K}^{(t)})^T}{\sqrt{D_k}} \right) \quad (6.5)$$

The final output for the temporal index t is the weighted sum of the Values based on the attention scores. A residual connection is employed to add the retrieved high-frequency information directly back to the original radar signal, facilitating stable gradient flow:

$$\mathbf{Z}^{(t)} = \mathbf{A}^{(t)} \mathbf{V}^{(t)} + \mathbf{X}^{(t)} \quad (6.6)$$

6.2.2 Software Implementation (Custom Layer)

To execute this mathematical translation within the MATLAB environment, the RadarECG-WaveletCrossAttentionLayer was authored. The implementation prioritizes vectorized windowing and dimensional stability to avoid the severe computational bottlenecks typically associated with local attention mechanisms.

```
classdef RadarECGWaveletCrossAttentionLayer < nnet.layer.Layer & nnet.layer.  
    Formattable  
    properties (Learnable)  
        Query  
        KeyWeights  
        KeyBias  
        ChannelEmbedding  
        ValueWeights  
        ValueBias  
    end  
  
    properties  
        NumRadarChannels  
        NumECGWavelets  
        KeyDimension  
        WindowRadius  
    end
```

```

methods
    function layer = RadarECGWaveletCrossAttentionLayer(numRadarChannels,
numECGWavelets, keyDimension, windowRadius, name)
        layer.Name = name;
        layer.Description = "Local Radar-to-ECG Wavelet Cross-Attention (
Vectorized)";
        layer.NumRadarChannels = numRadarChannels;
        layer.NumECGWavelets = numECGWavelets;
        layer.KeyDimension = keyDimension;
        layer.WindowRadius = windowRadius;

        % Learnable Parameters initialization
        layer.Query = randn(numECGWavelets, keyDimension, 'single') *
sqrt(2/keyDimension);
        layer.KeyWeights = randn(keyDimension, 1, 'single') * sqrt(2);
        layer.KeyBias = zeros(keyDimension, 1, 'single');
        layer.ChannelEmbedding = randn(keyDimension, numRadarChannels, '
single') * 0.05;
        layer.ValueWeights = randn(1, 1, 'single') * 0.1;
        layer.ValueBias = zeros(1, 1, 'single');
    end

    function Z = predict(layer, X)
        if isa(X, 'dlarray')
            fmt = dims(X);
            X = stripdims(X);
        else
            fmt = '';
        end

        orig_sz = size(X);
        C = orig_sz(1);
        T = orig_sz(2);

        if C ~= layer.NumRadarChannels
            error("Expected %d radar wavelet channels, received %d.",
layer.NumRadarChannels, C);
        end

        X_3D = reshape(X, C, T, []);
        B = size(X_3D, 3);

        R = layer.WindowRadius;
        K = layer.KeyDimension;
        E = layer.NumECGWavelets;
        W = 2 * R + 1;
        N = C * W;

```

```

% 1. BULLETPROOF VECTORIZED WINDOWING
Zpad = zeros(C, R, B, 'like', X_3D);
X_pad = cat(2, Zpad, X_3D, Zpad);
X_windows = zeros(C, W, T, B, 'like', X_3D);

token_idx = 1;
for offset = -R:R
    start_idx = R + 1 - offset;
    window_slice = X_pad(:, start_idx : start_idx+T-1, :);
    X_windows(:, token_idx, :, :) = reshape(window_slice, [C, 1,
T, B]);
    token_idx = token_idx + 1;
end

X_windows_perm = permute(X_windows, [2, 1, 3, 4]);
X_flat = reshape(X_windows_perm, N, T, B);

% 2. COMPUTE VALUES & KEYS
values = layer.ValueWeights(1,1) .* X_flat + layer.ValueBias(1,1)
;
keys_base = reshape(layer.KeyWeights, [K, 1, 1, 1]) .* reshape(
X_flat, [1, N, T, B]);
CE = repelem(layer.ChannelEmbedding, 1, W);
keys = keys_base + reshape(layer.KeyBias, [K, 1, 1, 1]) + reshape
(CE, [K, N, 1, 1]);

% 3. COMPUTE ATTENTION SCORES
keys_mat = reshape(keys, K, []);
scores_mat = (layer.Query * keys_mat) ./ sqrt(single(K));
scores = reshape(scores_mat, [E, N, T, B]);

% Softmax (Manual for Overflow avoidance)
maxScores = max(scores, [], 2);
expScores = exp(scores - maxScores);
attention = expScores ./ sum(expScores, 2);

% 4. WEIGHTED SUM & RESIDUAL CONNECTION
weighted_values = attention .* reshape(values, [1, N, T, B]);
out = sum(weighted_values, 2);
Z_3D = reshape(out, [E, T, B]) + X_3D;

out_sz = orig_sz;
out_sz(1) = E;
Z = reshape(Z_3D, out_sz);

if ~isempty(fmt)
    Z = dlarray(Z, fmt);
end

```

```

        end
    end
end

```

LISTING 6.2: Implementation of the Local Radar-to-ECG Wavelet Cross-Attention Layer.

The algorithmic execution ensures that the local attention mechanism scales efficiently across the extended 2048-point sequences:

- **Vectorized Window Extraction:** Rather than processing each temporal position independently, the implementation extracts all local windows using shifted, zero-padded tensor operations, reducing explicit iteration over the sequence length. Instead, the layer utilizes zero-padding and extracts the entire sequence shifted by specific offsets. By concatenating these shifted tensors, the local context (Keys and Values) is across all temporal indices simultaneously, enabling massive parallelization.
- **Dimensional Flattening and Broadcasting:** To apply the dense attention operations, all trailing dimensions (such as Batch size) are dynamically collapsed into a unified 3D tensor using the reshape function. This guarantees that dot products between the globally learned Queries and the local Keys execute deterministically without dimension mismatch errors.
- **Numerically Stable Softmax:** Exponentiating raw attention scores can easily lead to numerical overflow, returning Infinity and collapsing the network’s gradients. The custom implementation actively subtracts the maximum score along the token dimension before applying the exponential function, ensuring strict numerical stability during backpropagation.
- **Residual Integration:** After the local Values are weighted by the attention mask and summed, they represent the retrieved high-frequency ECG motif. This output is directly added to the original un-flattened tensor. This additive bypass inherently stabilizes deep networks and allows the architecture to strictly focus on learning the missing morphological peaks rather than attempting to reconstruct the entire underlying baseline from scratch.

While the proposed Local Cross-Attention mechanism bridges the architectural gap by shifting from temporal attenuation to cross-domain translation, it introduces a critical hardware trade-off: **high memory consumption**.

Because the algorithm constructs explicit localized spatio-temporal windows ($\mathbf{X}_{windows}$ of size $C \times W \times T \times B$) and projects them into expanded key-value spaces ($\mathbf{K}$ of size $K \times N \times T \times B$),

the intermediate activation tensors scale aggressively with sequence length T and batch size B . For instance, operating with a batch size of $B = 450$ over a sequence length of $T = 2048$ points causes these vectorized tensors to instantaneously consume several gigabytes of active memory during the backpropagation phase.

Although full vectorization successfully eliminates explicit MATLAB for-loops and drastically accelerates execution time, it trades computational speed for a massive RAM/VRAM footprint. Consequently, this architectural choice imposes a strict hardware constraint: training requires careful tuning of the batch size to prevent Out-Of-Memory (OOM) crashes, serving as a practical reminder of the engineering challenges when scaling transformer-inspired attention mechanisms to long-sequence biomedical time series.

6.2.3 Experimental Results & Evaluation

The Local Cross-Attention architecture was trained and evaluated under the standard baseline conditions (Apnea, Resting, Valsalva). As recorded in the training logs (Figure 6.4), the model successfully completed 200 epochs, but the computational cost was severe, extending the training duration to approximately 1 hour and 1 minute. This empirically confirms the theoretical complexity trade-off analyzed in Section 6.2.2, demonstrating a massive escalation in the memory footprint and processing overhead due to vectorized windowing and expanded key-value tensors.

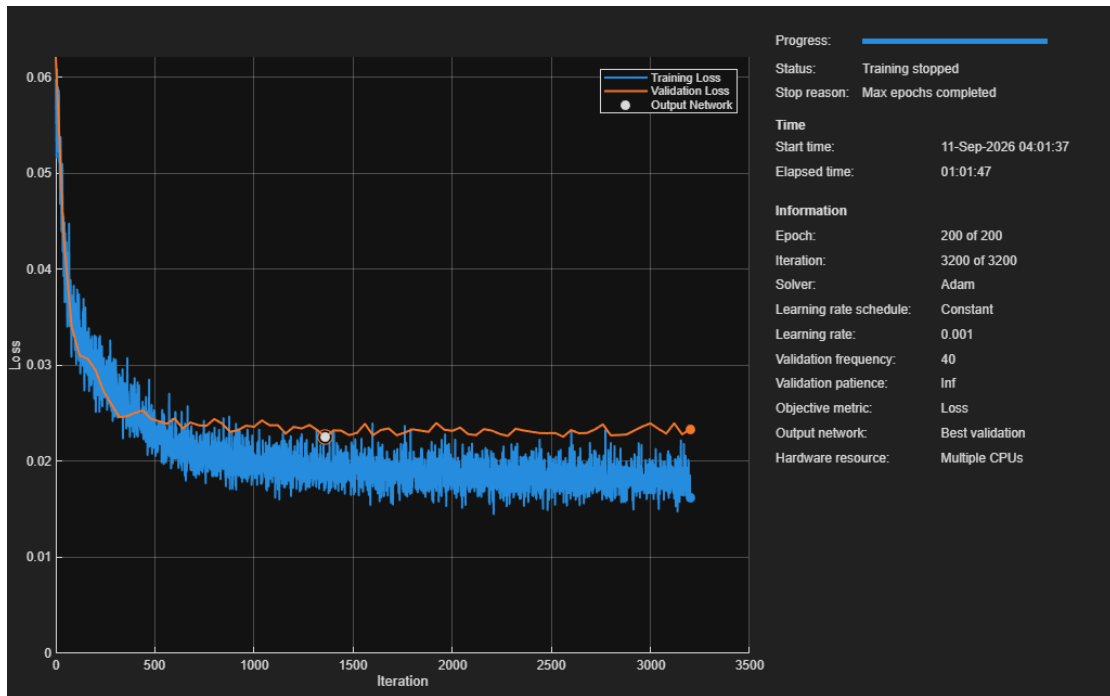
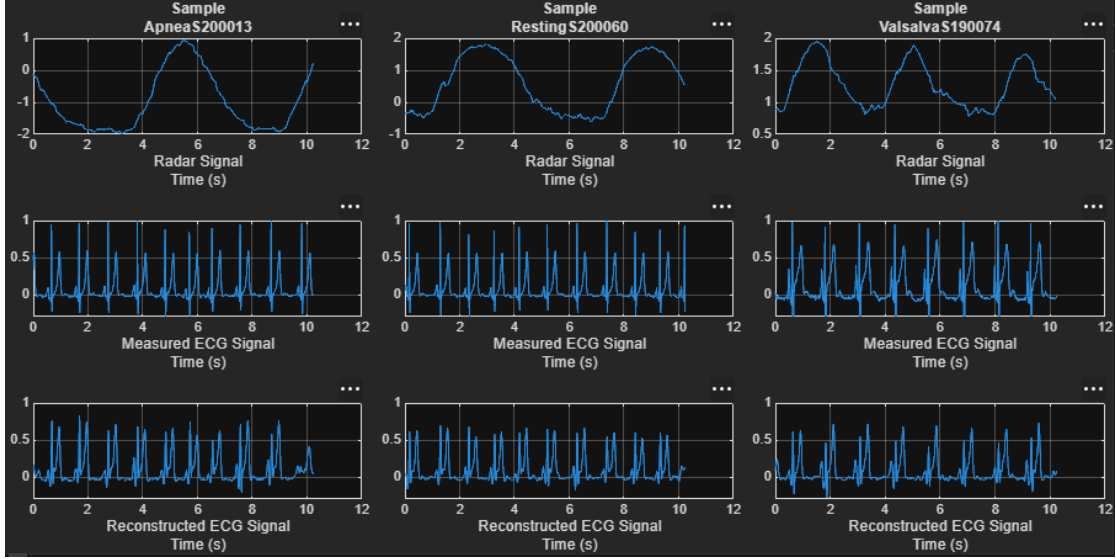


FIGURE 6.4: Training and validation loss convergence for the Local Cross-Attention architecture, highlighting an execution time exceeding one hour.

Quantitatively, the cross-attention mechanism achieved a Mean Test MSE of 0.043109. Compared to the original baseline of 0.04374, this represents a negligible quantitative reduction of approximately 1.44%. To evaluate whether this tiny numerical shift reflects actual morphological recovery, Figure 6.5 contrasts the baseline model directly against the Cross-Attention output.

Baseline Architecture (No Cross-Attention)



Experimental Architecture (With Local Cross-Attention)

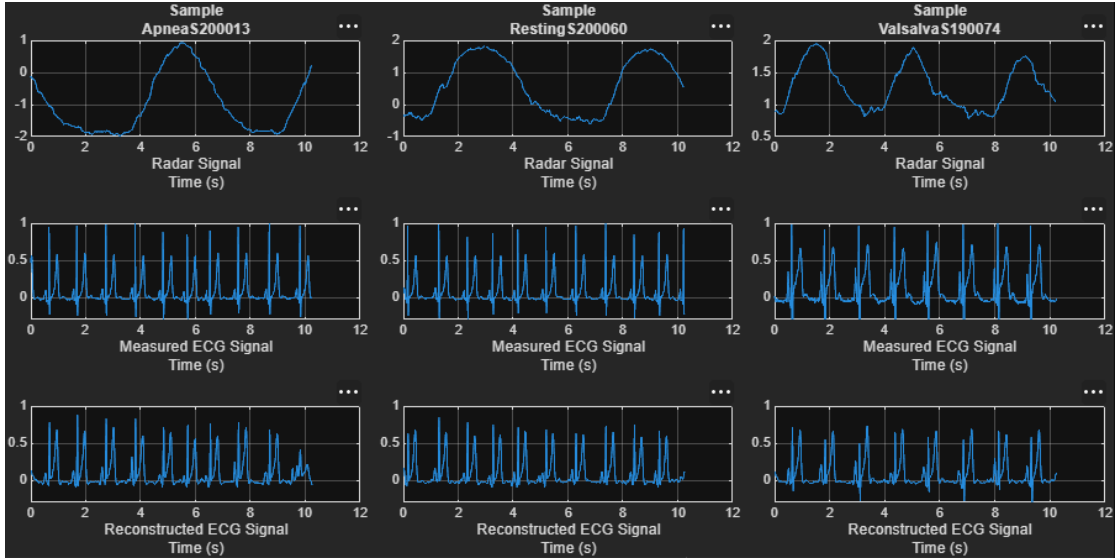


FIGURE 6.5: Direct visual comparison between the baseline model (top) and the Local Cross-Attention model (bottom). Morphological flattening of the QRS complexes persists with negligible visual differences.

A rigorous visual inspection of Figure 6.5 reveals that the anticipated cross-domain translation does not materialize in practice. While minor pixel-level variations can be discerned, the

structural limitations remain completely unresolved: the network still fails to synthesize true high-amplitude R-peaks, and the morphological flattening persists.

Critical Scientific Assessment: This experiment yields a vital negative conclusion for the scope of this thesis. Although the theoretical premise—shifting from a spatial filter to a cross-attention translator—is conceptually sound, the results demonstrate that increasing architectural complexity does not inherently bring the system closer to solving the root problem. The Local Cross-Attention model imposes an unacceptable hardware penalty (roughly tripling the training time and severely burdening system memory) for an almost non-existent performance return (1.44% MSE drop).

Ultimately, this experiment proves that while advanced localized attention mechanisms can extract deeper structural interactions, they are inefficient and practically unviable for contact-less radar-to-ECG reconstruction. The data confirms that more detailed mathematical modeling does not automatically bypass the underlying physical barriers of the modality, providing a clear boundary for future architectural explorations.

6.3 Limits of Spatial Synthesis: Cross-Scale Morphological Fusion

Given the significant computational bottleneck and limited morphological recovery observed in the Local Cross-Attention experiment, the final architectural exploration shifts toward a zero-parameter, physics-informed approach. Rather than relying on the network to independently discover complex temporal or cross-domain translations, this section proposes a direct mathematical synthesis of the desired features.

The core hypothesis is built upon the principles of Wavelet Shrinkage and Mathematical Morphology. The preceding Channel Attention layer successfully scales the multiresolution channels based on their Signal-to-Noise Ratio (SNR). However, instead of passing all five scaled channels forward, this experiment introduces a hard pruning mechanism. The two lowest-ranked channels—typically representing high-frequency baseline noise—are entirely discarded. The remaining three dominant channels are then mathematically fused using non-linear morphological operations (dilation and erosion) to explicitly synthesize the missing high-frequency transients (peaks and valleys) before feeding the tensor into the convolutional decoder.

6.3.1 Mathematical Formulation of Morphological Fusion

Let $\mathbf{X}$ represent the weighted multiresolution tensor outputted by the Channel Attention module. Based on the intrinsic physical properties of the MODWT decomposition for radar-derived

vital signs, levels 1 and 2 predominantly capture structural and atmospheric noise, while levels 3, 4, and 5 encapsulate the biological harmonics (respiration and cardiac motion).

Consequently, the hard pruning operation drops levels 1 and 2, explicitly isolating the dominant scales:

$$W_{winner} = \mathbf{X}_4, \quad W_{second} = \mathbf{X}_5, \quad W_{third} = \mathbf{X}_3 \quad (6.7)$$

To synthesize the missing transient features without introducing learnable parameters, morphological interaction is enforced across these dominant scales. A synthesized peak channel (C_{peak}) is constructed by extracting the element-wise maximum (constructive interference) between the two strongest wavelets. Conversely, a synthesized valley channel (C_{valley}) is constructed by extracting the element-wise minimum (destructive interference) between the winner and the third wavelet:

$$C_{peak}(t) = \max(W_{winner}(t), W_{second}(t)) \quad (6.8)$$

$$C_{valley}(t) = \min(W_{winner}(t), W_{third}(t)) \quad (6.9)$$

To maintain structural compatibility with the downstream convolutional network (which expects 5 input channels), the final output tensor $\mathbf{Z}$ is formed by concatenating the three raw dominant channels with the two newly synthesized morphological channels:

$$\mathbf{Z} = [W_{winner}; W_{second}; W_{third}; C_{peak}; C_{valley}] \quad (6.10)$$

6.3.2 Software Implementation (Custom Layer)

To seamlessly integrate this mechanism, the MorphologicalSynthesisLayer was developed in MATLAB. Because this layer operates entirely on element-wise morphological arithmetic, it requires no learnable parameters, executing with an optimal temporal complexity.

```
classdef MorphologicalSynthesisLayer < nnet.layer.Layer & nnet.layer.  
    Formattable  
    methods  
        function layer = MorphologicalSynthesisLayer(name)  
            layer.Name = name;  
            layer.Description = "Cross-Scale Morphological Fusion (Max/Min)";  
        end  
  
        function Z = predict(~, X)  
            if isa(X, 'dlarray')  
                fmt = dims(X);  
                X_data = stripdims(X);  
            else
```

```

        fmt = '';
        X_data = X;
    end

    sz = size(X_data);
    cDim = find(sz == 5, 1);

    if isempty(cDim)
        error("Dimension error! Expected a dimension of size 5 (
Wavelets).");
    end

    idx = repmat({' ':'}, 1, ndims(X_data));

    idx{cDim} = 4; W_winner = X_data(idx{:});
    idx{cDim} = 5; W_second = X_data(idx{:});
    idx{cDim} = 3; W_third = X_data(idx{:});

    C_peak = max(W_winner, W_second);
    C_valley = min(W_winner, W_third);

    Z_data = cat(cDim, W_winner, W_second, W_third, C_peak, C_valley)
;

    if ~isempty(fmt)
        Z = dldarray(Z_data, fmt);
    else
        Z = Z_data;
    end
end
end
end
end

```

LISTING 6.3: Implementation of the Cross-Scale Morphological Synthesis Layer with dynamic dimension resolution.

The algorithmic execution ensures that the tensor dimensions remain completely stable across the forward pass without incurring computational overhead:

- **Deterministic Pruning:** By directly indexing the physically relevant MODWT levels via dynamic dimension scanning, the layer bypasses dynamic sorting functions. This guarantees a smooth, uninterrupted gradient flow during backpropagation, avoiding the severe graph discontinuities associated with dynamic index selection.
- **Vectorized Element-wise Morphology:** The mathematical synthesis of the peak and valley channels relies strictly on the native max and min operators. These vectorized

functions execute instantaneously across the entire sequence length and batch dimension, completely neutralizing the computational bottlenecks observed in the previous attention-based architectures.

- **Channel Conservation:** The concatenation (cat) along the appropriate axis perfectly restores the depth of the tensor to exactly 5 channels. This allows the custom layer to act as a modular drop-in replacement, requiring zero structural modifications to the input dimensions of the subsequent 1D convolutions.

6.3.3 Experimental Results & Evaluation

The Morphological Fusion network demonstrated exceptional computational efficiency during training. As recorded in the training logs (Figure 6.6), the model successfully completed 200 epochs in approximately 9.5 minutes. This zero-parameter, $O(T)$ complexity approach completely eliminated the severe hardware bottleneck introduced by the local cross-attention mechanism in the previous experiment.

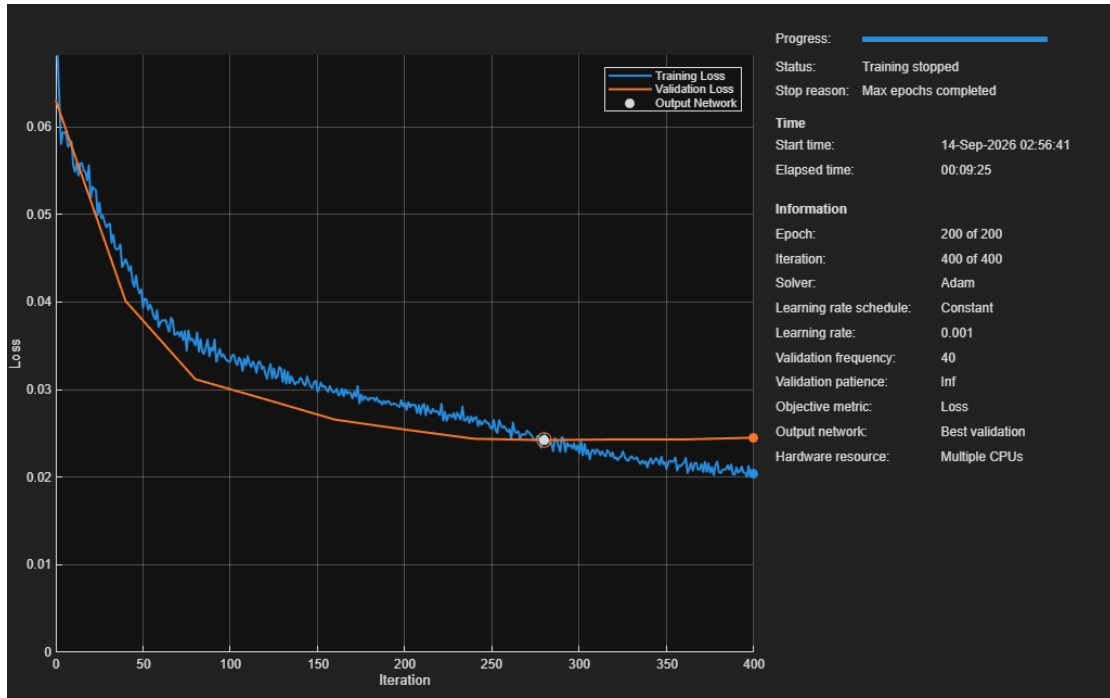
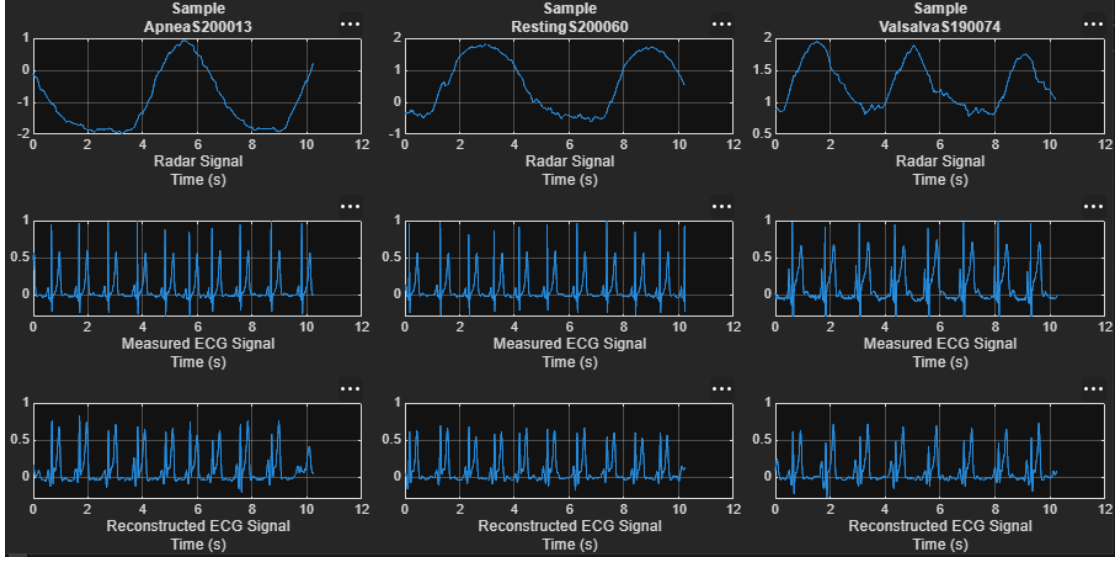


FIGURE 6.6: Training and validation loss convergence for the Morphological Fusion architecture, demonstrating rapid execution time (9.5 minutes) and a highly stable validation curve.

Quantitatively, the model achieved a Mean Test MSE of **0.03785**. Compared to the original baseline MSE of 0.04374, this represents a substantial error reduction of **13.46%**—a massive numerical improvement over the negligible $\sim 1.4\%$ changes seen in the attention-based experiments. However, a rigorous visual inspection of the reconstructed signals (Figure 6.7) reveals a

critical discrepancy between the mathematical optimization of the MSE metric and the actual clinical objective of this thesis.

Baseline Architecture



Experimental Architecture (Morphological Fusion)

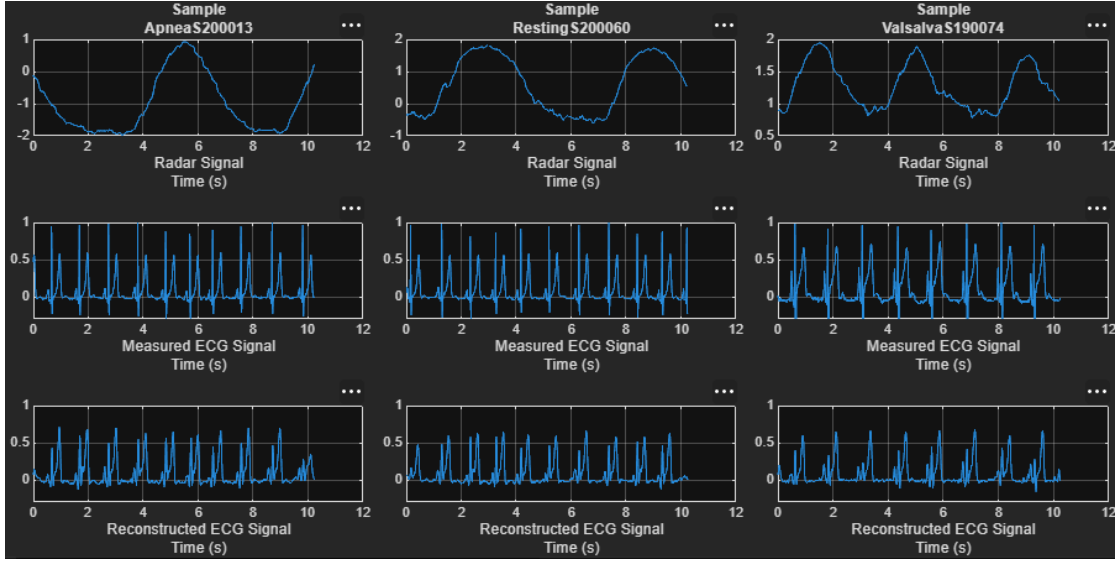


FIGURE 6.7: Visual comparison between the baseline and Morphological Fusion models. While inter-beat noise is reduced, the QRS complexes fail to sharpen and appear artificially broadened.

A critical analysis of the morphological reconstruction exposes the following realities regarding the experiment's true performance:

- **The MSE Illusion (Baseline Cleaning):** The 13.46% drop in the Mean Squared Error is highly misleading regarding R-peak reconstruction. The improvement is primarily attributed to the hard pruning of the lower MODWT scales (levels 1 and 2). By eliminating

high-frequency baseline artifacts, the network produced much cleaner inter-beat intervals. Since these flat intervals account for the vast majority of the time-domain signal, cleaning them drastically improves the overall mathematical error, masking the model's failure at the peaks.

- **Failure of Sharp Transient Synthesis:** Despite the improved metric, the primary objective—reconstructing the sharp, vertical R-peaks—was not met. The constructive interference via the max-operator forced the network to combine two dominant, inherently low-frequency wavelets. Mathematically, taking the maximum of two broad waves results in a bulkier, wider envelope. Consequently, instead of synthesizing a sharp high-frequency transient (a "needle"), the network generated artificially broadened, blunt peaks that deviate further from the true electrical ECG morphology.
- **The Need for Morphological Evaluation Metrics:** This experiment proves that relying solely on global regression metrics like MSE is inadequate for clinical signal translation. Future evaluations must incorporate morphology-aware metrics (e.g., Dynamic Time Warping, specialized peak-detection loss functions, or perceptual losses) that explicitly penalize the absence of high-amplitude transients rather than just average temporal deviations.

Architectural vs. Physical Limits

The culmination of these three experiments provides a deeply nuanced perspective on the challenge of radar-to-ECG translation. Experiments 1 and 2 proved that naively increasing spatial and temporal complexity (via attention mechanisms) fails to synthesize missing high-frequency transients and merely overloads system memory. Experiment 3 proved that forcing the network through physics-informed morphological synthesis can successfully clean the signal and optimize standard loss functions, but it inherently fails to recover the sharp electrical R-peak, often broadening it instead.

It is crucial to emphasize that this persistent morphological flattening does not automatically imply that the radar sensor completely lacks the necessary physiological information. While it is highly probable that macroscopic mechanical chest vibrations are fundamentally too smooth to contain micro-transient electrical fidelity, the bottleneck may equally lie within the current deep learning paradigms. Traditional 1D-CNNs, even when augmented with advanced signal processing layers, act primarily as sophisticated filters and interpolators.

Synthesizing a sharp electrical spike from a mechanical wave may ultimately require moving beyond deterministic signal processing and regression architectures. Future research must explore advanced generative deep learning paradigms—such as Generative Adversarial Networks (GANs), Diffusion Models, or Transformers guided by custom morphological loss functions—capable of learning the non-linear physiological mapping between mechanical motion

and electrical depolarization. The challenge is immense, but the intersection of advanced deep learning and raw sensor data remains an open and highly promising frontier.

Chapter 7

Conclusions and Future Work

7.1 Summary of the Thesis

This thesis investigated the feasibility of translating mechanical chest displacements, captured using Doppler radar, into continuous Electrocardiogram (ECG) waveforms. The central motivation was to examine whether deep learning can bridge the modality gap between mechanical manifestations of cardiac activity and the true electrical activity represented by the ECG.

The study followed a progressive architectural methodology. The initial baseline utilized a 1D-CNN framework. This was subsequently extended through Temporal Attention to capture temporal dependencies, and Local Cross-Attention to model localized interactions between the radar-derived representation and the target ECG. Finally, a physics-informed Morphological Fusion layer based on the Maximal Overlap Discrete Wavelet Transform (MODWT) was developed to explicitly synthesize the wavelet components associated with distinct morphological characteristics.

Each experiment was designed to address a specific limitation identified in the preceding approach. Overall, the experiments demonstrated that the proposed framework can reconstruct meaningful large-scale waveform characteristics and substantially suppress inter-beat noise. However, the reconstruction of fine-grained ECG morphology, particularly the sharp high-frequency characteristics of the QRS complex, remained consistently challenging across all architectures.

7.2 Main Findings and Final Conclusions

The experiments conducted throughout this thesis reveal a crucial distinction between general waveform reconstruction and accurate morphological synthesis. Table 7.1 summarizes the main experimental stages and their principal outcomes.

TABLE 7.1: Summary of the main architectural experiments and their observed outcomes.

Architecture	Test MSE	Change	Main Observation
Baseline 1D-CNN	0.04374	–	Established the reference performance for radar-to-ECG reconstruction.
Temporal Attention	0.04313	–1.39%	Produced a small reduction in MSE without a meaningful improvement in ECG morphology.
Local Cross-Attention	0.04311	–1.44%	Provided a similar MSE improvement at a substantially higher computational cost.
Morphological Fusion	0.03785	–13.46%	Significantly reduced global MSE and noise, but produced broadened rather than sharply reconstructed QRS complexes.

7.2.1 The Discrepancy Between MSE and Morphological Fidelity

The Morphological Fusion experiment revealed one of the most important findings of this thesis: an improvement in a conventional regression metric (MSE) does not necessarily correspond to an improvement in physiological fidelity. While the model achieved a substantial 13.46% reduction in MSE, qualitative inspection demonstrated that this improvement was primarily driven by the suppression of inter-beat noise. The R-peaks remained considerably broader and less sharply defined than those in the reference ECG. Because long inter-beat intervals occupy a substantial proportion of the time-domain signal, minimizing noise in these regions disproportionately improves the global MSE, masking the model’s inability to synthesize short-duration, high-amplitude electrical transients.

7.2.2 Architectural Limitations and Physical Constraints

The persistent flattening of the R-peaks across all experiments highlights a challenging reconstruction bottleneck. Increasing architectural complexity (Attention mechanisms) or forcing physics-informed operations (Morphological Fusion) failed to perfectly synthesize the sharp QRS morphology.

However, it is vital not to interpret these findings as definitive proof of a fundamental physical impossibility. The results prove that deterministic regression architectures, guided by global MSE, are insufficient for this highly non-linear translation. While the mechanical radar signal is inherently smoother than the electrical ECG, alternative learning objectives and generative approaches may potentially uncover and exploit physiological correlations that were not effectively utilized by the methods investigated here.

7.3 Limitations of the Study

Several experimental limitations should be considered when interpreting these results:

- **Evaluation Metrics:** The primary quantitative evaluation relied on Mean Squared Error. As demonstrated, MSE is inadequate for evaluating sharp morphological features. The absence of morphology-aware metrics (e.g., Dynamic Time Warping, R-peak amplitude error) limits the extent to which true physiological reconstruction can be objectively quantified.
- **Subject-Independent Generalization:** The dataset consists of 30 healthy volunteers. If samples from the same subjects exist in both training and testing sets, the performance may partially reflect subject-specific characteristics. A strict subject-wise validation protocol, such as leave-one-subject-out validation, would provide a more rigorous assessment of subject-independent generalization.
- **Healthy-Subject Dataset:** The conclusions cannot be extended to pathological ECG morphologies. The ability to reconstruct waveforms in healthy subjects does not establish diagnostic capability for cardiovascular diseases.
- **Restricted Experimental Conditions:** Architectural comparisons were primarily evaluated under stable conditions (Apnea, Resting, Valsalva). Highly non-stationary conditions involving severe body movement (e.g., Tilt-Up, Tilt-Down) introduce complex artifacts that require further investigation.

7.4 Future Work

Future progress in radar-to-ECG translation should evolve beyond deterministic regression architectures.

7.4.1 Morphology-Aware Loss Functions

Training objectives should explicitly emphasize clinical morphology. A composite loss function incorporating terms for R-peak localization, QRS width, and waveform correlation could force the network to preserve short-duration high-amplitude events that contribute little to the global MSE but are paramount for ECG validity.

7.4.2 Reinforcement Learning for Morphological Preservation

Waveform reconstruction could be formulated as a sequential optimization problem using Reinforcement Learning (RL). An RL agent could be guided by a custom reward function that explicitly penalizes broad, low-frequency envelopes while heavily rewarding the synthesis of sharp QRS complexes. Recent literature, such as the 2026 study by the City University of Hong Kong [1], successfully demonstrated the combination of RL and State-Space Modeling for radar-based ECG reconstruction, providing a relevant direction for addressing the R-peak flattening observed in this thesis.

7.4.3 Advanced Generative Modeling and Sequence Architectures

Generative Adversarial Networks (GANs) and Diffusion Models could be leveraged to learn the underlying distribution of realistic ECG morphologies, actively synthesizing sharper high-frequency structures from smooth mechanical inputs. Furthermore, full Transformer-based architectures—evaluated strictly against their computational overhead—could provide multi-head attention over the entire sequence, potentially capturing long-range physiological interactions more effectively than CNNs.

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